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Bounty hunter optimizer: A novel metaheuristic with an application to multi-UAV mobile edge computing and path planning

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ABSTRACT

This study proposes the bounty hunter optimizer (BHO), a novel optimization algorithm inspired by the behavior of bounty hunters. Unlike traditional methods that rely on mean aggregation, BHO employs a decentralized position-update strategy based on local differences and random disturbances, inherently avoiding center bias and population collapse. BHO introduces three core innovations: (1) the Explorpolis rule, which quantifies individual fitness and its spatial relationship to the high-quality solution set, enabling intelligent filtering of inferior solutions; (2) a quantum probability rotation selection mechanism, which replaces the traditional roulette-wheel selection using quantum bits and rotation gates to continuously adjust probability amplitudes, significantly enhancing population diversity; and (3) an evolutionary model inspired by the "thorough investigation," "rough search," and "hunter re-assignment" behaviors of hunters, integrated with a self-feedback adjustment mechanism to dynamically balance exploration and exploitation. Experimental results on the CEC2017 test suite demonstrate that the BHO excels in both 30-dimensional and 100-dimensional optimization tasks, significantly outperforming 11 advanced optimization algorithms, including four competition-level state-of-the-art variants. When applied to the multi-unmanned aerial vehicle (multi-UAV) mobile edge computing (MEC) and path planning (PP) (multi-UAV MEC-PP) system, BHO effectively addresses the path optimization problem under quality-of-service constraints. Extensive simulations across different scales confirm its superior performance, achieving >15% cost reduction relative to advanced competitors and providing reliable and efficient support for PP and decision-making in MEC environments.

1. Introduction

High-dimensional constrained optimization problems are widely encountered in real-world engineering systems, where decision variables are tightly coupled and feasibility is restricted by multiple operational constraints. In general, algorithms for solving such problems can be broadly categorized into deterministic and stochastic approaches [1, 2]. Deterministic methods are often effective for convex formulations and can approximate optimal solutions through gradient-based updates or progressive refinement of the feasible domain; representative examples include the steepest descent method, the Lagrange method, and branch-and-bound schemes [3]. However, many practical problems exhibit strong nonconvexity and complex structural characteristics, under which deterministic methods may become trapped in suboptimal local solutions and struggle to obtain high-quality global solutions. By contrast, metaheuristic (MH) algorithms, as an important branch of stochastic methods, introduce randomness into the search process and thus enhance global exploration capability without requiring restrictive modeling assumptions [4]. MH algorithms typically start from randomized solutions and iteratively exploit historical search feedback to explore the solution space, often yielding effective and reliable high-quality solutions [5–7].

Ever since the last century, many researchers have been dedicated to the development and improvement of MH algorithms. One of the early

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achievements is represented by genetic algorithm (GA) [8]. Inspired by natural selection, GA uses objective function values as fitness measures to simulate biological reproduction, crossover, and mutation, continually evolving individuals in the population to gradually approach the optimal solution. Later, the well-known particle swarm optimization (PSO) [9] algorithm emerged, inspired by bird flocking behavior: during the search process, "particles" share position and food information, naturally converging toward regions with high-quality solutions, thereby improving solving efficiency. The emergence of such MH algorithms laid the foundation for solving more complex, higher-dimensional, and multi-objective coupled optimization problems, driving further development and innovation in MH algorithms [10,11].

In addition to GA and PSO, some MH algorithms that simulate natural physical phenomena have attracted considerable attention in recent years. Notable examples include simulated annealing (SA) [12] and Black Hole Algorithm (BHA) [13]. The SA algorithm is inspired by the physical process wherein metal materials gradually cool from a high-energy state to a low-energy stable state. It generates new solutions by introducing random perturbations and gradually reduces the probability of accepting inferior solutions on the basis of the Metropolis criterion, eventually approaching the global optimum. The BHA algorithm simulates the phenomenon of a black hole's strong gravitational pull attracting the surrounding matter. During the iteration process, the best solution in the current population is regarded as the "black hole" and the positions of other solutions are dynamically updated based on the distance and gravitational relationship. Solutions sufficiently close to the black hole are absorbed and considered its "satellites," thereby continuously moving closer to the global optimum.

Since 2010, more biomimetic MH algorithms have emerged [12], such as Whale Optimization Algorithm (WOA) [14], Grey Wolf Optimizer (GWO) [15], and Sparrow Search Algorithm (SSA) [16]. WOA simulates the hunting behaviors of humpback whales, including encircling, spiral attacks, and random searching, effectively enhancing global search capabilities. GWO locates prey through the α, β, and δ wolves and guides the Ω wolf in tracking it, thereby achieving rapid convergence. SSA simulates social interactions associated with foraging and defense through the division of roles into discoverers, joiners, and scouts, efficiently solving optimization problems in the search space. Unlike PSO, which often converges prematurely into local optima, these biomimetic MH algorithms better handle complex, high-dimensional, and multi-objective problems by diversifying search phases and role assignments [17,18].

Although biomimetic algorithms have improved global search capabilities relative to earlier classical methods, their development faces a significant bottleneck—center bias [19]. Typical methods such as GWO and WOA update the population using a few high-quality individuals and centralized aggregated guidance (e.g., explicit mean/centroid attraction or leader-weighted combinations); this can statistically introduce a systematic drift toward a local attraction basin. Consequently, the population may collapse toward a local "central" region and lose diversity. Meanwhile, parameter sensitivity is a common issue [20]. For instance, the learning factors in PSO and the pheromone evaporation rate in ant colony optimization significantly affect optimization efficiency, and there exists a lack of consistent configurations across different problems [21]. Furthermore, the commonly used roulette-wheel selection (RWS) employs discrete probability sampling and is highly sensitive to weight scales. When a few elite weights dominate, they excessively increase selection pressure, intensifying population concentration and decreasing diversity. When individual weights are similar, sampling variance significantly increases. Both factors hinder diversity maintenance and robust convergence. The convergence of center bias, parameter sensitivity, and RWS sampling effects in high-dimensional, non-convex spaces significantly increases the risk of premature failure. This synergy often leads to solution collapse and makes it difficult to satisfy complex engineering constraints.

To overcome these intertwined issues, this study proposes a novel MH optimization algorithm—the bounty hunter optimizer (BHO). Notably, BHO draws inspiration from the social behavior of bounty hunter organizations in pursuit of prey. It gradually approaches the target by adaptively adjusting the search area and the movement range of individuals (hunters), significantly reducing the use of fixed parameters. Unlike mean/centroid aggregation guidance (i.e., attraction toward a population mean/centroid or other weighted-average center), BHO does not compute or follow any aggregated center. Instead, BHO uses the personal, group, or global best as best-so-far exemplar solutions (rather than averages); combined with local-difference perturbations and explicit Explorpolis set filtering, this design largely avoids center bias and reduces the risk of premature population collapse. To address the discrete sampling and scale sensitivity issues of RWS, this study introduces the quantum probability rotation selection (QPRS) mechanism as an alternative. By continuously representing preferences with amplitude and smoothly updating in small steps, it achieves low variance and adjustable selection pressure, strengthening cross-group information flow and diversity maintenance. In implementation, BHO employs clustering initialization and group construction to improve coverage and avoid redundant sampling. It introduces a region value evaluation based on "bounty scores" to adaptively schedule individual behavior and group strategies. Meanwhile, region representative points and high-quality solutions are solely used as directional feedback to update the attraction coefficients (rather than mean aggregation terms), thus achieving a dynamic balance between exploration and exploitation while maintaining structural diversity in the search.

To further confirm the applicability of BHO in practical engineering optimization, this study constructs a multi-UAV mobile edge computing and path planning (multi-UAV MEC-PP) model with explicit QoS constraints as the verification object. MEC brings computation, storage, and networking capabilities closer to the data source at the edge to support low-latency and reliable services [22]; it has been widely studied as a key enabler for edge intelligence and latency-sensitive applications [23,24]. Deploying UAVs as aerial edge nodes improves coverage and service agility. However, their mobility introduces complex interdependencies between task allocation, wireless connectivity, and trajectory constraints, such as obstacle avoidance. This results in a large-scale, highly non-convex, and tightly constrained joint optimization problem [25–27]. Therefore, jointly optimizing task allocation and PP for multi-UAV collaboration remains a challenging problem that deserves further investigation.

In recent years, UAV-enabled MEC optimization has been investigated using multiple paradigms, including model-based optimization (e.g., decomposition/alternating optimization and convex-approximation techniques) and learning-based methods for dynamic environments [28–30]. For example, in a previous study [31], a multi-agent strategy based on Q-learning was employed to optimize UAV flight paths, combined with echo state networks to predict the mobile trajectory of end-users; Liu et al. [32] proposed a QoS evaluation model and a reward maximization strategy. As practical deployments require strict compliance with flight dynamics and obstacle constraints, recent studies have advanced beyond generic metaheuristics by explicitly incorporating motion planning and constrained trajectory design into joint UAV-MEC optimization. Representative efforts follow several distinct paths. Some studies employ large-scale formulations that couple discrete association decisions with continuous trajectory variables [33]. Others use decomposition-based designs, applying ADMM for resource allocation alongside successive convex approximation for trajectory optimization [34]. Additionally, learning–control hybrids have emerged, where deep reinforcement learning generates reference trajectories that are subsequently enforced by classical LQR-based trackers [35]. For obstacle-rich missions, sampling-based planners, such as an improved bidirectional RRT, have been employed to efficiently generate collision-free trajectories for MEC-driven operations [36]; similar ideas have been extended

to disaster-rescue settings through multi-agent frameworks that jointly optimize obstacle avoidance, coverage, and task offloading [37]. A recent survey further summarized these trajectory-aware offloading strategies and their associated coupling mechanisms, offering a consolidated view of the corresponding trade-offs [38]. While these studies establish a strong foundation, the search space expands exponentially in more complex scenarios involving numerous end nodes, dense obstacles, and stringent QoS constraints [39,40]. The existing methods still face convergence difficulties or fail to simultaneously meet all constraints in high-dimensional joint UAV-MEC optimization.

To address these limitations, we propose BHO as a joint optimization framework for the multi-UAV MEC-PP problem. Under the premise of satisfying QoS constraints, the paths of each UAV are jointly planned to minimize the overall system cost. We set up two sets of simulation comparison experiments, ranging from low- to high-dimensional scenarios. The results demonstrate that BHO achieves more stable convergence and a lower objective value in this engineering scenario, showing its optimization advantages under complex constraints. The following are the main contributions of this study:

- This study draws inspiration from the behavioral characteristics of bounty hunters in pursuing targets and proposes BHO. The algorithm constructs a swarm intelligence framework without mean aggregation and centralization operators, thereby inherently suppressing center bias, maintaining diversity, and enabling robust global search for high-dimensional nonconvex problems.
- We introduce the Explorpolis set-layer filtering criterion, which utilizes a non-threshold acceptance rule to selectively retain complementary solutions. This mechanism effectively suppresses population homogenization, thereby enhancing convergence stability and overall solution quality.
- The QPRS mechanism is developed to replace traditional RWS. By employing probability amplitudes and rotational updates, QPRS provides smooth, adjustable selection pressure that mitigates weight-scale sensitivity and high variance, reducing the risk of premature convergence.
- On the CEC2017 test suite (30D and 100D), BHO achieves significant advantages in overall metrics, outperforming mainstream and competition-winning algorithms. The statistical significance of the differences between BHO and the comparison methods was assessed using the Wilcoxon rank-sum test.
- BHO is applied to the multi-UAV MEC-PP task with QoS constraints. Comparative simulations across various scales show that BHO yields lower system costs and smoother convergence, highlighting its robustness and practical potential for complex, highly constrained engineering optimization.

The structure of the remaining part of the study is as follows: Section 2 introduces the inspiration behind and mathematical model of the BHO algorithm and analyzes its algorithmic complexity. Section 3 presents the comparison of the algorithms and the experimental setup. Section 4 experimentally confirms the optimization performance of BHO using the CEC2017 test suite. Section 5 constructs a multi-UAV MEC-PP model and performs two sets of simulation comparison experiments. Section 6 provides conclusions and outlook, summarizing the entire study and discussing future research directions.

2. Proposed bounty hunter optimizer

The inspiration for BHO comes from the pursuit process of bounty hunter organizations—intelligence gathering, task deployment, collaborative division of labor, and capture. To avoid the center bias and population collapse common in mean-driven algorithms, BHO adheres to three core principles:

1) **Decentralized updates**, which utilize local differences and random disturbances rather than mean operators.
2) **Structured diversity maintenance**, employing set-layer acceptance to prevent homogenization.
3) **Adjustable selection pressure**, using smooth probability updates to mitigate the high variance inherent in discrete roulette-wheel selection.

These principles are mapped to four modules in the algorithm: (1) K-means initial construction, using multiple clusters to enhance early hit rates and reduce redundancy; (2) A dynamic division of labor is adopted, with fine-grained search conducted within groups and coarse-grained search performed between groups, transitioning gradually from exploration to exploitation over the course of iterations; (3) Explorpolis set-layer filtering, using the "fitness $\times$ normalized distance" criterion to retain complementary structures and suppress homogenization; (4) QPRS achieves continuous sampling for cross-group coordination, balancing information flow and selection intensity. Subsequent sections will provide the mathematical formulations and complexity analysis.

2.1. Mathematical model of BHO

2.1.1. Oversampling initialization (Intelligence gathering)

Inspired by the information-collection stage of bounty-hunter organizations, BHO begins with an oversampling initialization step that aims to provide broad coverage of the feasible search space.

In the mathematical formulation, each candidate solution is a D-dimensional decision vector, where each dimension corresponds to a decision variable. For a population size N, we generate an oversampled candidate pool X_o (e.g., $10N$samples) within the feasible bounds, and construct it as follows:

$$X_o = \begin{vmatrix} X_{o_1} \\ X_{o_2} \\ \vdots \\ X_{o_{10N}} \end{vmatrix} = \begin{vmatrix} x_{o_{1,1}} & x_{o_{1,2}} & \cdots & x_{o_{1,D}} \\ x_{o_{2,1}} & x_{o_{2,2}} & \cdots & x_{o_{2,D}} \\ \vdots & \vdots & \ddots & \vdots \\ x_{o_{10N,1}} & x_{o_{10N,2}} & \cdots & x_{o_{10N,D}} \end{vmatrix} \tag{1}$$

where, D denotes the position of clues in the multidimensional space, that is, the dimensionality of the optimization problem. X_o represents the clue positions obtained by the bounty hunter organization through net-casting at ten times the number of hunters, expressed as a $10N \times D$ matrix. The factor 10 was empirically chosen to ensure sufficient initial sampling coverage without incurring excessive computational cost.

2.1.2. Clustering-based representative selection and grouping (Task deployment)

After generating the oversampled candidate set, we select a representative and well-dispersed initial population to avoid redundant initialization in the same local region of the search space. Since all sampled candidates are a priori equally plausible before evaluation, directly using random samples may lead to multiple individuals being initialized in a highly similar neighborhood. To obtain a diverse and structured initialization, we apply K-means clustering to the oversampled set X_o. By minimizing intra-cluster variance while encouraging separation among clusters, K-means partitions X_o into N clusters. The resulting cluster centroids serve as representative candidates of their neighborhoods and are sufficiently separated from each other, which helps maintain population diversity at the start of the search. We set the number of clusters to N and use the N centroids as the initial population X, defined as follows:

$$X = \begin{vmatrix} X_1 \\ X_2 \\ \vdots \\ X_N \end{vmatrix} = \begin{vmatrix} x_{1,1} & x_{1,2} & \cdots & x_{1,D} \\ x_{2,1} & x_{2,2} & \cdots & x_{2,D} \\ \vdots & \vdots & \ddots & \vdots \\ x_{N,1} & x_{N,2} & \cdots & x_{N,D} \end{vmatrix} \tag{2}$$

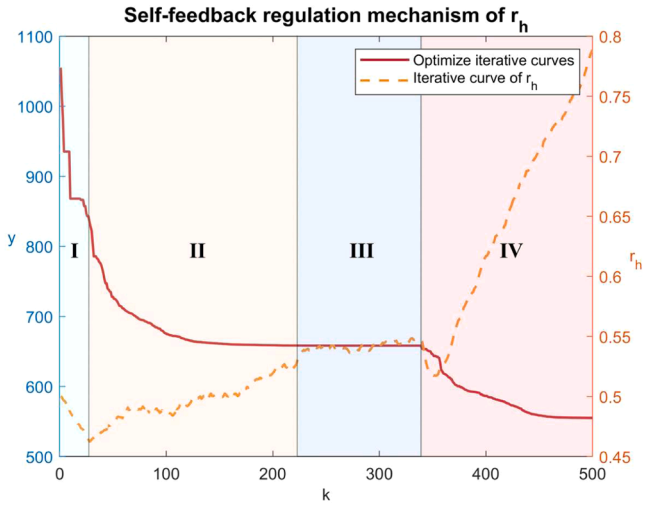


Fig. 1. Self-feedback regulation mechanism of r_h.

Once the dispatch positions of the bounty hunters are determined, the organization divides them into pursuit groups based on the distances between their assigned positions and arranges for them to act independently. BHO employs the K-means algorithm to simulate the organization's grouping behavior through clustering, dividing the bounty hunters into N_{group} groups.

2.1.3. Adaptive operator scheduling (Collaborative division of labor)

Each individual adaptively selects its search operator based on its relative contribution/rank within the subgroup. Intuitively, individuals with better performance focus on exploitation (local refinement) in their neighborhood, while those with poorer performance emphasize exploration to enlarge the search range and reduce redundant effort in unpromising regions.

In BHO, the search mode of the i-th individual at iteration $k$is controlled by a binary indicator $s_i^k \in \{0,1\}$, where $s_i^k = 1$ denotes the exploitation (refinement) operator and $s_i^k = 0$ denotes the exploration operator. Let $R_{h,i}^k$ denote the rank of individual $i$within its subgroup at iteration k (with 1 being the best), and let r_h^kbe the threshold used to determine the search mode. Then,

$$s_i^k = 1\{R_{hi}{}^k \le r_h^k\} \tag{3}$$

When $s_i^k = 1$, the hunter performs thorough investigation; otherwise, it performs rough search.

It can be observed that the value of r_h directly determines the proportion of hunters adopting the two update strategies in the current generation of the organization. To enhance exploitation when clues are abundant and to increase the proportion of exploration-mode hunters when progress stagnates, this paper adopts a self-feedback mechanism to update r_h, which is revised using the following formula:

$$r_h^k = \begin{cases} 0.5, if\ k = 1 \\ r_h^{k-1} + \sum_{i \in M_t} score_i^{k-1} - \sum_{i \in M_r} score_i^{k-1}, if\ k > 1 \end{cases} \tag{4}$$

where, M_t denotes the set of hunters operating in the thorough investigation mode; M_r denotes the set of hunters operating in the rough search mode; *score* represents the update result of the current hunter, a positive score indicates performance improvement, while a negative score reflects stagnation or degradation. The following equation describes the score updating process:

$$score_i^k = \begin{cases} -\dfrac{1}{N * k_{max}}, F_i^k \le F_i^{k-1} \\ \dfrac{1}{N * k_{max}}, F_i^k > F_i^{k-1} \end{cases} \tag{5}$$

where, the k_{max} is the Maximum number of iterations

When progress stagnates and the algorithm enters a locally convergent state, the scores of hunters performing thorough investigation are often negative, causing r_h to decrease and thereby enhancing the exploration ability of the algorithm. Conversely, when the algorithm discovers a new optimal clue and the hunters performing thorough investigation move closer to this solution, their scores usually become positive, r_h gradually increases, and the exploitation ability of the algorithm is enhanced. This realizes the self-feedback adjustment of BHO based on optimization results. During one optimization process, Fig. 1 illustrates the variation curve of r_h in the BHO algorithm.

As shown in Fig. 1, the self-feedback threshold r_h roughly divides the evolution of BHO into four phases: (1) During the early exploration stage, clues superior to random initialization are quickly obtained, leading to a slight decrease in r_h; (2) As exploration exhausts the available information, the hit rate of high-quality clues declines, while the marginal benefit of thorough investigation becomes more prominent, causing r_h to gradually increase; (3) When thorough investigation enters a diminishing returns phase and early high-quality clues are exhausted, individuals cluster around the current optimum with little improvement, causing r_h to fluctuate and fall back to around 0.5, which increases the proportion of exploration; (4) Once the optimum is surpassed, thorough investigation expands around the new optimal clue and frequently updates results, while large-scale random searches in the later stage yield fewer hits. Consequently, r_h rises sharply, enabling the algorithm to escape local stagnation and approach the global optimum.

Thorough investigation. Individuals assigned to the exploitation mode correspond to the top-ranked members of the population/subgroup. These individuals update their positions by leveraging three best-reference solutions: the current global best, the subgroup best, and their personal best. By jointly using these well-defined reference targets, the exploitation mode emphasizes local refinement around high-quality regions, typically with a relatively small effective step size. The update rule is given by:

$$\begin{aligned} X_{i,j}^k test &= X_{i,j}^k + m_i^k \\ &* \left(\begin{array}{c} c_{1_i}^k * rand * \left(X_{best,j}^k - X_{i,j}^k\right) + c_{2_i}^k * rand * \left(X_{best,j}^{k,team_l} - X_{i,j}^k\right) + \\ c_{3_i}^k * rand * \left(X_{i,j}^{best} - X_{i,j}^k\right) \end{array} \right) \end{aligned} \tag{6}$$

where, X_{best}^k denotes the position of the highest-valued clue at iteration k, and $X_{best,j}^k$ represents its value in dimension j; $X_{best}^{k,team_l}$ denotes the position of the highest-valued clue within group l at iteration k, and $X_{best,j}^{k,team_l}$ represents its value in dimension j; X_i^k denotes the position of hunter i at iteration k, where $X_{i,j}^k$ is its value in dimension j. $X_{i,j}^k test$ represents the value of hunter i in dimension j after evolution; $c_{1_i}^k$, $c_{2_i}^k$, and $c_{3_i}^k$ are the reference factors of hunter i with respect to the global best clue, the group best clue, and its own best clue, respectively, each updated within the range [0,1]; m_i^k is the movement factor of hunter i at iteration k, also updated within the range [0,1]. Its specific update rule is given later in Eq. (18).

During the thorough investigation phase, the reference factors c_1, c_2, and c_3 are adaptive reference weights (analogous to PSO learning factors) that control the influence of the global/subgroup/personal best references. They are initialized randomly in $[0,1]$and updated after each exploitation step according to the individual's improvement magnitude as follows:

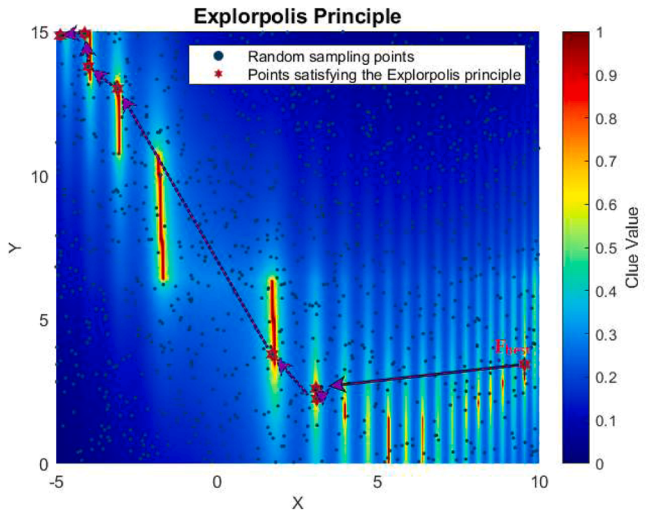


Fig. 2. Explorpolis rule.

$$c_i^k = \begin{cases} \min\left\{c_i^{k-1} + \frac{R_{hi}^k}{10}, 0.8\right\}, if\ c_i^k \geq \overline{c_i^k}\ and\ R_{hi}^k \leq 0.2 \\ \max\left\{c_i^{k-1} - \frac{R_{hi}^k}{10}, 0.2\right\}, if\ c_i^k < \overline{c_i^k}\ and\ R_{hi}^k \leq 0.2 \\ c_i^k \quad , if\ 0.2 < R_{hi}^k < 0.8 \\ \min\left\{c_i^{k-1} + \frac{R_{hi}^k}{10}, 0.8\right\}, if\ c_i^k < \overline{c_i^k}\ and\ R_{hi}^k \geq 0.8 \\ \max\left\{c_i^{k-1} - \frac{R_{hi}^k}{10}, 0.2\right\}, if\ c_i^k \geq \overline{c_i^k}\ and\ R_{hi}^k \geq 0.8 \end{cases} \tag{7}$$

where, $\overline{c_i^k}$ represents the mean of the three reference factors, defined as $\overline{c_i^k} = \frac{c_{1_i}^k + c_{2_i}^k + c_{3_i}^k}{3}$. After a thorough investigation, the obtained clue achieves a high ranking in value, the weight configuration of c_1, c_2, and c_3 is regarded as effective. In this case, the BHO algorithm preserves the relative magnitudes of the individual reference factors and slightly amplifies their differences; otherwise, it significantly reverses and adjusts the reference factors.

Rough search. Individuals with relatively poor performance (lower contribution/rank) are assigned to the exploration mode. In this mode, an individual does not rely on explicit best-reference targets; instead, it samples a random reference solution from its subgroup (or neighborhood) to guide a coarse search step. If the sampled reference yields better fitness than the individual's current solution, the individual moves toward the reference; otherwise, it moves away to promote diversification. Since exploration-mode individuals lack strong target guidance, they typically employ a larger effective step size together with random perturbations to enlarge the covered region and reduce redundant search in unpromising areas. The update rule is expressed as follows:

$$X_{i,j}^k rough = \begin{cases} X_{i,j}^k + rand * \left(X_{r,j}^{k,team_l} - X_{i,j}^k\right), if\ F_i^k \leq F_r^{k,team_l} \\ X_{i,j}^k - rand * \left(X_{r,j}^{k,team_l} - X_{i,j}^k\right), if\ F_i^k > F_r^{k,team_l} \end{cases} \tag{8}$$

where, $X_r^{k,team_l}$ denotes the random clue position within search group l at iteration k, and $X_{r,j}^{k,team_l}$ represents its value in dimension j; $X_i^k rough$ denotes the position that hunter i investigates through inquiry at iteration k, and $X_{i,j}^k rough$ represents its value in dimension j; F_i^k represents the clue function value currently obtained by hunter i, while $F_r^{k,team_l}$ denotes the value of the random clue within group l.

After reaching the inquiry-determined investigation position, such hunters conduct a random search within a larger radius. The corresponding update process is expressed by the following equation:

$$X_{i,j}^k test = \begin{cases} X_{i,j}^k rough + m_i^k * rand_1 * X_{i,j}^k rough, if\ rand_2 > 0.5 \\ X_{i,j}^k rough - m_i^k * rand_1 * X_{i,j}^k rough, if\ rand_2 \leq 0.5 \end{cases} \tag{9}$$

After obtaining new clues $X_i^k test$ in the current generation, if their value is superior to the corresponding clue from the previous generation, the new clue is directly retained. If it is not superior, the organization merges the new and old clues and applies the innovative Explorpolis rule for selection.

Explorpolis rule. The Explorpolis rule specifies that a candidate solution is accepted if no solution in the elite set simultaneously outperforms it in both distance and fitness. The distance between any candidate solution and an elite solution is calculated as follows:

$$TD_i^k = \frac{\sqrt{\sum_{j=1}^{D}\left(\frac{X_{new,j}^{k+1} - X_{i,j}^k}{U_j - L_j}\right)^2}}{D} \tag{10}$$

where, X_{new}^{k+1} denotes the updated or selected elite solution, and $X_{new,j}^k$ represents its value in dimension j; U is the upper bound of the problem in all dimensions, and U_j is its value in dimension j; L is the lower bound of the problem in all dimensions, and L_j is its value in dimension j.

The Explorpolis rule arranges clues in descending order of value, sequentially admitting solutions that are not eliminated, and uses the admitted solutions to filter subsequent candidates. Let $S\{x_i\}$ denote the set of admitted elite solutions, x_{test} the candidate solution, and x_i the current solution. The normalized nearest distance from the candidate solution to the elite set is defined as:

$$TD(X_{test}, S) = \min_{x_i \in S} \frac{X_{test} - X_{old}}{U - L}_2 \tag{11}$$

Under the minimization setting, given the elite set $S\{x_i\}$, the current solution X_{test}, and the candidate solution X_{old}, the normalized nearest distance TD from the candidate solution to the elite set is calculated according to Eq. (11). elite set S is updated dynamically after each generation based on current best solutions. The acceptance logic is as follows: first, if the objective function value of the candidate solution is better than that of the current solution, the candidate solution is directly accepted and replaces the current solution; Otherwise, the elite set is checked for any solution that is both better than the candidate and within the distance threshold τ. If no such "better-and-closer" solution exists in the elite set, the candidate solution is accepted to enrich population diversity; otherwise, the candidate is rejected and the current solution is retained. This criterion preserves candidate solutions only when they provide performance improvement or structural complementarity, thereby suppressing homogenization and centralized clustering.

In this study, the clue values were generated as the background using a two-dimensional single-objective test function: $z = \left(y - \frac{5.4}{4\pi^2}x^2 + \frac{5}{\pi}x - 6\right)^2 + 10\left(1 - \frac{1}{8\pi}\right)\cos(x^2) + 10.$ Subsequently, 1000 solutions were randomly generated in the search space, and the Explorpolis rule was applied to filter them. The resulting outcomes are illustrated in Fig. 2.

As shown in Fig. 2, the Explorpolis rule begins with the current best solution and examines candidate solutions sequentially in descending order of fitness. For each candidate, its normalized nearest distance to the admitted elite set is first computed. If the candidate outperforms the current solution, it is directly accepted; Otherwise, the candidate is admitted only if no solution in the elite set simultaneously has fitness not worse than the candidate and lies within a radius τ of it. This prevents homogenized stacking and redundant replacements, thereby producing a more representative and complementary solution set. Unlike the

Table 1
Quantum rotation angle update rules.

	$Q_{best,l}$	$C^{k,l} > C^{k,best}$	$\alpha_l\beta_l > 0$	$\alpha_l\beta_l < 0$	$\alpha_l = 0$	$\beta_l = 0$
0	0	False	0	0	0	0
0	0	True	0	0	0	0
0	1	False	$+0.01\pi$	-0.01π	0	$\pm0.01\pi$
0	1	True	-0.01π	$+0.01\pi$	$\pm0.01\pi$	0
1	0	False	-0.01π	$+0.01\pi$	$\pm0.01\pi$	0
1	0	True	$+0.01\pi$	-0.01π	0	$\pm0.01\pi$
1	1	False	0	0	0	0
1	1	True	0	0	0	0

widely used Metropolis criterion [41], Explorpolis does not rely on probabilistic thresholds such as temperature or annealing schedules. Instead, it employs a single geometric radius τ as a diversity constraint to balance exploitation and exploration, thereby enhancing the global search and generalization ability of BHO while reducing the risk of local convergence.

2.1.4. Intensification and QPRS-based reassignment (capture)

As the pursuit progresses, the organization narrows down the target based on the gathered intelligence and gradually reduces the hunters' search range. Once the target appears, the pursuit group takes immediate and decisive action to ensure the target is surrounded and captured before it can escape.

Hunter reassignment. When an individual consistently ranks near the bottom of its subgroup for multiple iterations, this indicates potential stagnation or poor suitability to the current search region. To mitigate prolonged unproductive search and promote information transfer, the algorithm triggers a reassignment step, relocating the individual to a more promising subgroup based on the group-quality selection mechanism.

BHO defines a threshold called Max Poor Performance Count ($MPPC \geq 2$)to identify hunters requiring reassignment. When a hunter has consecutively ranked in the bottom 20% of the population for a number of times equal to this threshold, BHO reassigns it to a superior pursuit group using QPRS, based on the groups' cumulative contributions. Its access privilege is temporarily elevated, allowing it to review the historical best clues of its new group and top-performing hunters, thereby enabling thorough investigation and accelerating adaptation to the new environment. The update equation for a reassigned hunter in this iteration is defined as follows:

$$X_{i,j}^{k}test = X_{i,j}^{k} + m_i^k * \left(\begin{array}{c} c_{1_i}^{k} * rand * \left(X_{best,j}^{k} - X_{i,j}^{k} \right) + c_{2_i}^{k} * rand * \left(X_{best,j}^{k.team_{new}} - X_{i,j}^{k} \right) \\ + c_{3_i}^{k} * rand * \left(X_{better,j}^{best.team_{new}} - X_{i,j}^{k} \right) \end{array} \right) \tag{12}$$

where, $X_{best}^{k.team_{new}}$ denotes the best clue within the new group after the hunter's reassignment, and $X_{best,j}^{k.team_{new}}$ represents its value in dimension j; $X_{better}^{best.team_{new}}$ denotes the historical best clue of another hunter in the new group whose performance is superior, and $X_{better,j}^{best.team_{new}}$ represents its value in dimension j.

Quantum Probability Rotation Selection. In QPRS, each reassignment group is represented by a quantum bit composed of two amplitude constants (α,β), which satisfy the following condition:

$$\alpha^2 + \beta^2 = 1 \tag{13}$$

The reassignment group is determined by measuring the quantum bit. When a uniformly distributed random number $r > \alpha^2$, the group is selected; otherwise, it is discarded. If more than one group is selected, repeated measurements are performed among the selected groups until only one group remains; If no group is selected, repeated measurements are carried out among the discarded groups until one group is chosen. The detailed procedure is presented in the pseudocode of Algorithm 1.

This mechanism allows quantum chromosomes with identical amplitude constants to yield different selection outcomes during measurement, thereby endowing the algorithm with significant flexibility and enhanced information-processing capability. Unlike traditional discrete selection, QPRS maintains a continuous probability state, allowing smooth transition between exploitation and exploration through small rotation-angle updates.

The initial amplitude constant of each group, $\alpha^{1,l}$, can be calculated using the following equation:

$$\alpha^{1,l} = \frac{C^{1,l}}{\sum C^{1,l}}, \tag{14}$$

where, $C^{k,l}$ denotes the cumulative contribution of group l at iteration k, defined as the sum of the cumulative contributions of its members, and it decays over iterations with a fading factor ρ. The specific calculation is given by:

$$C^{k,l} = \begin{cases} 0, & if\ k = 0 \\ (1-\rho) * C^{k-1,l} + \sum_{i \in B} c_i^k, & if\ k > 0 \end{cases} \tag{15}$$

where, ρ is the contribution fading factor, with its value constrained within [0,1]; B denotes the set of hunters in pursuit group l; c_i^k represents the contribution of hunter i at iteration k, and it is computed as follows:

$$c_i^k = \begin{cases} 0, if\ k = 0 \\ 1 - \frac{F_i^k}{\sum F_i^k}, if\ k > 0 \end{cases} \tag{16}$$

The update of the amplitude constant relies on the quantum rotation gate $U_\beta(\theta)$, and the specific computation is given by:

$$\begin{bmatrix} \alpha' \\ \beta' \end{bmatrix} = U_\beta(\theta) \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} \cos(\theta) & \sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} \tag{17}$$

where, $U_\beta(\theta)$ denotes the quantum rotation angle, which can be obtained from Table 1.

In the Table 1, Q denotes the group selection state (1 = selected, 0 = discarded); Q_l represents the selection state of group l; $Q_{best,l}$ denotes the selection state of the group with the highest contribution. To prevent non-convergent loops in rare cases, a finite-step guard ($guard_{MAX}$) is introduced during the QPRS random collapse process. If a team has not been selected after reaching this limit, a team is randomly chosen as the final output. This modification does not alter the expected selection distribution but improves the numerical robustness of the algorithm.

Encirclement. As better solutions accumulate, the population focuses on promising regions and the search shifts to intensification. The reassignment mechanism further redirects effort toward high-quality regions, accelerating this shift.

BHO employs iteration-dependent parameters to balance exploration and exploitation. Early stages prioritize global coverage, while later iterations focus on refining solution quality through intensified exploitation. Accordingly, as the iteration index increases, the algorithm gradually decreases the effective step size, resulting in a reduced search radius. The schedule is defined as follows:

$$m_i^k = 1 - \frac{k}{2k_{max}} \times \frac{F_i^k}{\sum F_i^k} \tag{18}$$

where, m_i^k denotes the movement factor of the i-th individual in the k-th iteration, which scales the step size in the update equation; F_i^k represents the objective function value of the i-th individual at iteration k, and $\sum F_i^k$

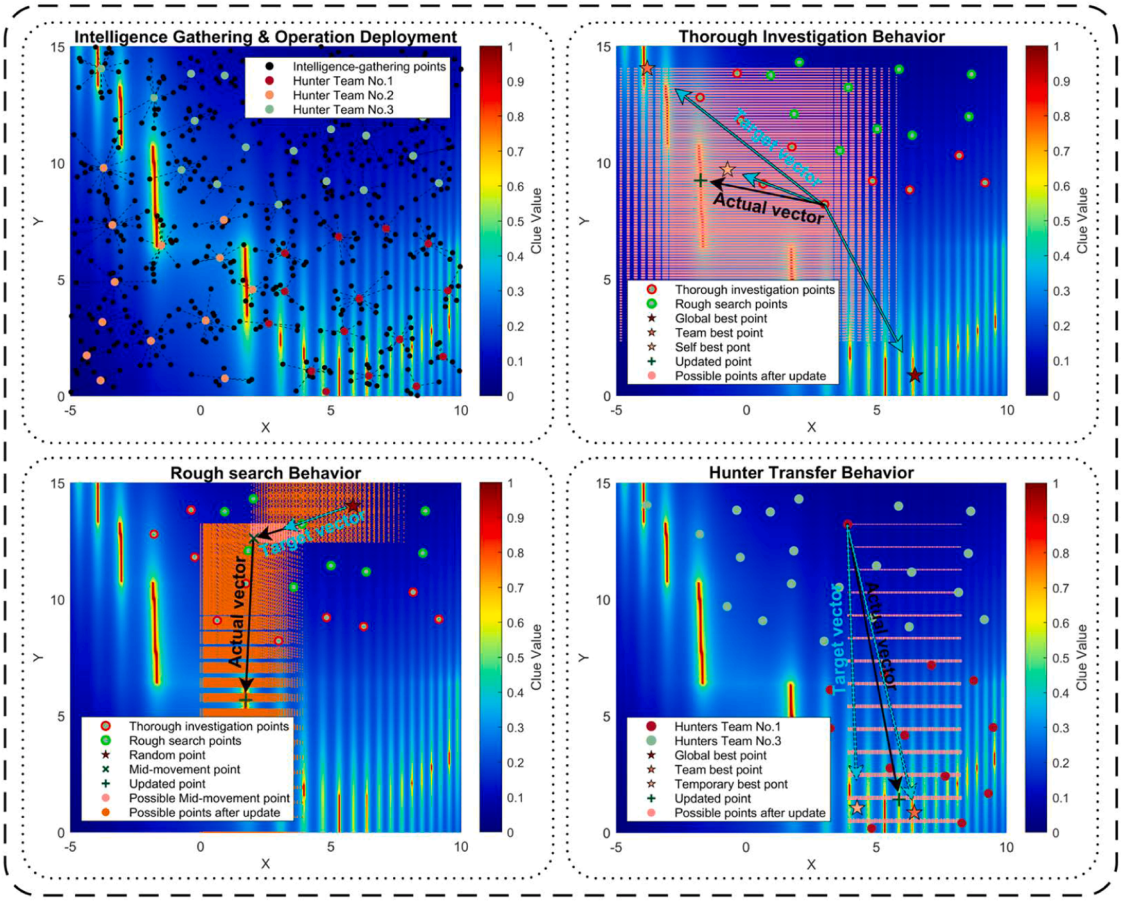


Fig. 3. Schematic diagram of BHO evolution.

Table 2
Parameters Definitions of BHO.

Parameter	Definition	Parameter	Definition
N	Population size (number of hunters)	$X_i^k rough$	position that hunter i investigates through inquiry at iteration k
D	Dimensionality of the optimization problem	$k,\ k_{max}$	Number of iterations and maximum number of iterations
U, L	Lower/upper bounds of the search space	X_i^k	Position of hunter i
X_o	Oversampled candidate pool	$X_{i,j}^k test$	Value of hunter i in dimension j after evolution
X	Initialized population (cluster centroids, $N \times D$)	X_{best}^k	Position of the highest-valued clue at iteration k
N_{group}	Number of pursuit groups (clusters for grouping)	P_i	Poor performance count of hunter i
s_i^k	Binary mode indicator (1: thorough investigation, 0: rough search)	$MPPC$	Max poor performance count triggering reassignment
$guard$, $guard_{MAX}$	Finite-step guard and its maximum to prevent non-convergent loops	$X_{better}^{best.team_{new}}$	Historical best clue of another hunter in the new group whose performance is superior
$R_{h_i}^k$	Rank of hunter $i$within its subgroup at iteration k	c_i^k	Contribution of hunter $i$at iteration k (for QPRS)
r_h^k	Threshold controlling mode switch (thorough investigation vs rough search)	$X_{best}^{k.team_{new}}$	Best clue within the new group after the hunter's reassignment
M_t, M_r	Sets of hunters in thorough investigation / rough search	ρ	Contribution fading factor in $C_{k,l}$ update
$score_i^k$	Self-feedback score for updating r_h	α_l, β_l	Qubit amplitudes for group l ($\alpha_l^2 + \beta_l^2 = 1$)
$C_{k,l}$	Cumulative contribution of group l at iteration k	F_i^k	Objective value of hunter $i$at iteration k
m_i^k	Movement factor (step-size scaling)	θ, $U_\beta(\theta)$	Quantum rotation angle and rotation gate used to update amplitudes
$c_{1_i}^k, c_{2_i}^k, c_{3_i}^k$	Reference factors for global/group/personal best (adaptive weights)	Q, Q_l, $Q_{\text{best},l}$	Group selection states used in QPRS update rule table
$\bar{c}_i^k$	Mean of the three reference factors $c_{1_i}^k, c_{2_i}^k, c_{3_i}^k$	φ_l	Selection status of team l in QPRS pseudocode
$X_r^{k.team_l}$	Random clue position within search group l at iteration k	N_s, N_{us}	Numbers of selected/unselected teams in QPRS
X_{new}^{k+1}	Updated or selected elite solution	TD_i^k	Normalized distance (used in Explorpolis)
S	Elite set used in Explorpolis filtering	τ	Radius parameter (diversity-preserving neighborhood threshold)

Table 3
Parameter settings of the chosen competitors.

Competitors	Name of the parameter	Value of the parameter	Years
GWO	a	[0,2]	2014
WOA	a, $a2$, b	[0,2], [- 1,−2], 1	2016
HHO	E_0, E_1	[- 1,1], [0,2]	2019
COA	C_1, C_2, μ, σ	0.2, 3, 25, 3	2023
IPO	w_1, w_2, α, P_1	0.9, 0.4, 1, 0.2	2024
MSMA	z, E,N	0.03, 100, 10	2023
OMA	m^r	[0,1]	2023
LSHADE_ cnEpSin	P, $\|A\|$, H	0.11, 1.4, 5	2017
LSHADE_SPACMA	p, $\|A\|$, H	0.11, 1.4, 5	2017
MadDE	p, $\|A\|$, H	0.18, 2.3, 10 × D	2021
EA4eig	h, $n0$, δ	4, 2 , 0.05	2022

denotes the sum of the objective values of all individuals in that iteration. Meanwhile, the reassignment mechanism gradually disbands pursuit groups with low yields, reallocating their hunters to more successful groups, thereby completing the encirclement of the target.

In summary, BHO constructs a fully decentralized swarm intelligence framework, integrating adaptive role division, quantum probability-based selection, and set-layer filtering, which jointly ensure convergence stability and population diversity across high-dimensional landscapes.

2.1.5. Overall description of BHO

Based on the aforementioned behavioral mechanisms, this paper presents the overall process and structure of the BHO. As shown in Fig. 3, BHO constructs an optimization framework coupling global exploration and local exploitation through a hierarchical process of intelligence gathering and task deployment - thorough investigation - rough search - hunter reassignment. All four subplots are displayed on a two-dimensional heatmap background, with the color bar on the right indicating the clue value. The top-left shows intelligence gathering and task deployment, where black sampling points are clustered by K-means into multiple initial groups (Team No.1–3), achieving global coverage. The top-right depicts thorough investigation, where green/red points represent fine/rough searches. Star markers are used to distinguish the global best, the team best, and the self-best point. “Target vector” and “Actual vector” indicate the theoretical and actual update directions, while the shaded bands represent potential landing positions after step-size shrinking. The bottom-left shows rough search, where random points and mid-movement points expand the search radius to cover unexplored regions. The bottom-right illustrates hunter reassignment, where different colored circles denote group members, and the “transfer vector” indicates the direction and updated position of cross-group migration. Together, these phases form a closed-loop optimization process. Updates are driven by local differences and disturbances, while the Explorpolis rule and QPRS maintain diversity. This synergetic design effectively suppresses center bias and ensures convergence toward the global optimum.

Additionally, Table 2 summarizes all symbols defined within the BHO, Algorithm 2 provides the pseudocode for BHO, offering a clear illustration of the entire execution process, from initialization and population division to behavioral updates and termination conditions, complementing the structure presented in Fig. 3.

2.2. Complexity analysis of the BHO

This section analyzes the time complexity of the BHO based on Algorithm 2. We examine the time complexity of each step in the algorithm and finally provide the overall complexity.

1. Initialize the algorithm parameters, N, k_{max}, D, N_{group}, ρ, L, U, c_1^0, c_2^0, c_3^0, r_h^1, t, C^0 : $O(1)$;
2. Randomization generates intelligence distribution points X_o : $O(1)$;
3. Kmeans clustering X_o get initial hunter position$X$: $O(10 \times N)$;
4. Kmeans clustering X groups: $O(N)$;
5. Calculate the target value for each hunter F : $O(k_{max} \times N)$;
6. Update the value scores of hunters c and hunter teams C : $O(k_{max} \times N)$;
7. Update individual, group, and collective optima X_i^{best}, $X_{best}^{k,l}$ and X_{best}^{k} : $O(k_{max} \times N)$
8. Update the hunter rank R_h, action ability m : $O(k_{max} \times N)$
9. Calculate the average of the reference factors for each hunter $\bar{c}$: $O(k_{max} \times N)$
10. Update the reference factors for each hunter c_1, c_2, c_3 : $O(k_{max} \times N)$
11. Deploy hunters who have performed poorly consecutively: $O(k_{max} \times N)$
12. Update the hunter's position according to the ranking$Xtest$: $O(k_{max} \times N)$;
13. Explorpolis rule of screening high quality solutions to determine the generation of hunter's final location X^{k+1} : $O(k_{max} \times N)$;
14. Update Hunter scores $score$: $O(k_{max} \times N)$;
15. Update the threshold for determining behavioral patterns r_h : $O(k_{max} \times N)$;
16. Number of update iterations k : $O(k_{max})$;
17. Correct the boundary of the solution: $O(k_{max})$;

Therefore, the optimizer-side overhead of BHO scales as $O(k_{max} \times N \times D)$ due to $O(D)$ coordinate-wise operations per individual per iteration; the objective-function evaluation cost is problem-specific and can be denoted as $C_f(D)$, giving a total runtime of $O\big(k_{\max} \times N \times \big(D + C_f(D)\big)\big)$. For fixed D, the optimizer-side term reduces to $O(k_{\max} \times N)$.

3. Experiment settings

All simulations were conducted on a Windows 11 platform with a 64-bit operating system. The experiments were implemented in MATLAB 2023b on a machine equipped with an AMD Ryzen 7 4800H CPU at 2.30 GHz and 16 GB RAM.

For performance comparison, several categories of algorithms were selected: (1) classical MH algorithms, including GWO, WOA, HHO [42]), recently proposed highly-cited algorithms IPO [43], MSMA [44]; (2) recently proposed and highly cited algorithms, including COA [45], OMA [46]; and (3) competition-level state-of-the-art algorithms with top CEC rankings, including LSHADE_SPACMA (CEC2017 winner) [47], LSHADE_cnEpSin (CEC2017 winner) [48], MadDE (CEC20221 winner) [49], EA4eig (CEC2022 winner) [50]. Table 3 summarizes the key parameter configurations of these algorithms, which are consistent with the original paper of the algorithms. The main control parameters of BHO are selected according to the sensitivity study in Section 4.1 (Fig. 4); in particular, we use $N_{\text{group}} = 4$, $\rho = 0.25$, $\tau = 0.3$, $r_h = 0.3$, which yields the best Friedman average rank.

In Section 4, the CEC2017 numerical optimization experiments were conducted with problem dimensions set to $D = 30$ and $D = 100$. The maximum function evaluations count was $500D$, population size was set to $N = 1D$, and each algorithm was independently executed 30 times.

In Section 5, BHO was further evaluated in the context of Multi-UAV MEC-PP. The small-scale scenario consisted of 10 obstacle nodes, 3 UAVs, and 6 terminal nodes, while the large-scale scenario included 40 obstacle nodes, 5 UAVs, and 20 terminal nodes. The problem dimension was defined as $D =$ Number of UAVs $\times$ 20 (with each UAV corresponding to 20 decision variables). Accordingly, the dimensions were $D = 60$ and $D = 100$ for the small- and large-scale cases, respectively. The maximum evaluation count was set to $400D$, the population size to $N = 1D$, and each algorithm was independently executed 5 times.

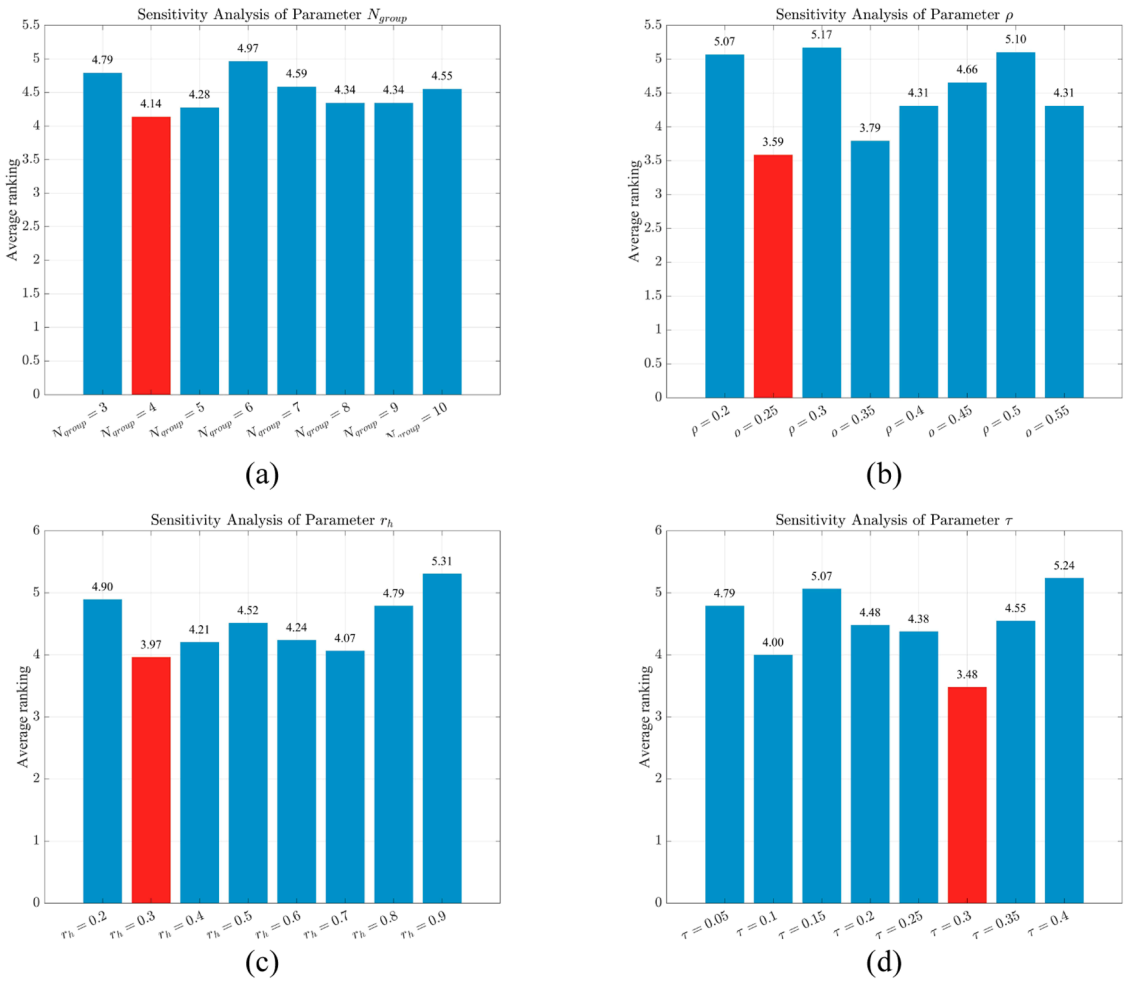


Fig. 4. Parameter sensitivity analysis of BHO.

4. Numerical optimization experiments and analysis

4.1. Parameter sensitivity analysis

To systematically investigate the influence of key control parameters on the overall performance of the proposed BHO, a parameter sensitivity analysis is conducted with respect to four core parameters, namely the number of groups N_{group}, the mode-switch threshold r_h, the contribution decay factor ρ in QPRS, and the radius parameter τ in the Explorpolis mechanism. These parameters respectively govern the group-wise population structure, the proportion between thorough and rough search behaviors, the temporal smoothing of inter-group selection pressure, and the diversity-preserving neighborhood around elite solutions. All sensitivity experiments are carried out on the CEC2017 benchmark suite with dimension $D = 30$. Except for the parameter under investigation, all experimental settings follow those in Section 3. Performance is evaluated using the Friedman average ranking over CEC2017 functions (lower is better), and a one-factor-at-a-time strategy is adopted to examine the effect of each parameter.

The experimental results are shown in Fig. 4. Overall, the performance differences among the tested parameter settings are not pronounced: the Friedman average rank changes within a relatively narrow range across the examined values. This indicates that the main control parameters have a limited impact on overall performance within reasonable intervals, and BHO does not rely on a highly specific, finely tuned configuration to achieve strong results. Therefore, Fig. 4 provides empirical evidence that BHO exhibits low parameter sensitivity and maintains stable performance under different parameter choices. For the group number N_{group}, the lowest Friedman average ranking (4.14) is obtained at $N_{group} = 4$, while both smaller and larger group settings lead to inferior performance. When N_{group} is too small (e.g., 3), the overall ranking deteriorates, and further increasing the group number (e.g., 6 and above) also results in performance degradation. For the contribution decay factor ρ, the best average ranking (3.59) is achieved at $\rho = 0.25$, whereas deviations from this value gradually worsen the ranking. Regarding the mode-switch threshold r_h, the lowest average ranking (3.97) is obtained when its initial value is set to 0.3, while relatively large values (e.g., 0.8 or 0.9) cause noticeable performance degradation. For the Explorpolis radius parameter τ, the optimal performance is achieved at $\tau = 0.3$, corresponding to the lowest average ranking of 3.48, whereas both excessively small and excessively large values lead to inferior rankings.

From a mechanistic perspective, these results reflect the distinct roles of the considered parameters in balancing exploration and exploitation. A small N_{group} restricts parallel exploration across different regions of the search space, whereas an overly large group number weakens within-group exploitation, making an intermediate group size more favorable. The decay factor ρ controls the forgetting rate of historical contribution information in QPRS: overly small values lead to sluggish selection responses, while overly large values may induce unstable selection pressure. Regarding r_h, lower initial values favor early-stage exploration to prevent premature convergence, while high values lead to excessive local intensification. Similarly, the radius τ regulates diversity in Explorpolis; extreme values—either too small or too large—can impair diversity maintenance and convergence speed. Moreover, although the Friedman rankings vary with different parameter settings,

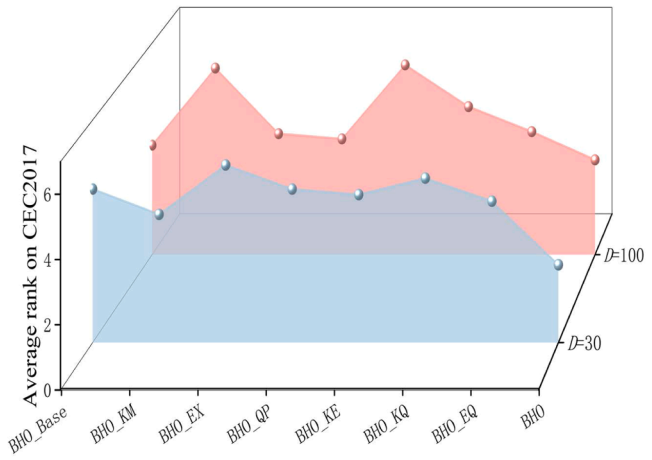


Fig. 5. Ablation study of BHO on CEC2017.

the performance differences remain moderate within most reasonable parameter ranges, and no abrupt degradation is observed. This indicates that the proposed BHO is not overly sensitive to precise parameter tuning and can maintain stable performance over a wide range of parameter values, thereby enhancing its robustness and practical applicability.

4.2. Ablation experiment

To systematically assess the respective contributions of the three proposed mechanisms, namely K-means initialization, Explorpolis, and QPRS, to BHO, we perform a comprehensive ablation study following two complementary protocols: progressive addition and progressive removal. Specifically, eight variants are considered: (i) BHO_Base, which disables all three mechanisms; (ii) BHO_KM, which adds only K-means initialization to the baseline; (iii) BHO_EX, which adds only Explorpolis; (iv) BHO_QP, which adds only QPRS; (v) BHO_KE, which combines K-means initialization and Explorpolis; (vi) BHO_KQ, which combines K-means initialization and QPRS; (vii) BHO_EQ, which combines Explorpolis and QPRS; and (viii) the full BHO, which integrates all three mechanisms. Experiments are performed on the CEC2017 benchmark suite with $D = 30$ and $D = 100$, and all other experimental settings follow those in Section 3. Fig. 5 summarizes the Friedman average rankings of the eight variants on CEC2017, where a smaller rank indicates better overall performance.

From Fig. 5, it can be observed that under $D = 30$, the full BHO has the lowest average ranking, which is significantly better than all pruned versions. In the single mechanism, BHO_KM has a clear improvement over BHO_Base, while BHO_QP is almost the same as BHO_base, and BHO_EX is obviously inferior to BHO_Base. For the two-mechanism variants, BHO_EQ provides a certain improvement relative to BHO_Base, BHO_KE only brings a small improvement, and BHO_KQ is worse than BHO_Base. For $D = 100$, the trend is more stringent: except for the full BHO, all single- and two-mechanism variants underperform BHO_Base, with the KM-involved variants (particularly BHO_KM and BHO_KE) showing the most pronounced degradation. The degradation of BHO_QP and BHO_EX is relatively light, but it still fails to stably surpass BHO_Base, and BHO_EQ does not form an advantage. It can be seen that in low dimensions, there is a "single point of visible benefit" (K-means initialization is the most prominent), but in high dimension, performance degrades once any mechanism is excluded, and only the complete mechanism combination has the consistent advantage across dimensions.

From the perspective of contribution and underlying mechanism, the three components regulate three key aspects of BHO: initial structure (K-means initialization), diversity preservation (Explorpolis), and selection-pressure scheduling (QP), whose effects are strongly complementary and coupled. In $D = 30$, K-means initialization directly improves early-stage coverage and group structure, thereby yielding a measurable benefit when enabled alone. However, without Explorpolis to suppress solution homogeneity and without QPRS to stabilize inter-

Table 4
Execution time comparison on CEC2017 (in seconds).

ID	GWO	WOA	HHO	IPO	MSMA	COA	OMA	LSHADE_cnEpSin	LSHADE_SPACMA	MadDE	EA4eig	BHO
F1	0.12	0.07	0.13	0.22	0.77	0.09	0.28	0.29	0.36	0.25	2.71	1.80
F2	0.12	0.06	0.13	0.10	0.73	0.08	0.27	0.25	0.19	0.23	2.60	1.65
F3	0.11	0.05	0.13	0.20	0.55	0.10	0.31	0.31	0.19	0.31	3.21	1.79
F4	0.12	0.05	0.13	0.19	0.65	0.07	0.28	0.24	0.17	0.15	2.04	1.43
F5	0.12	0.06	0.17	0.17	0.88	0.09	0.28	0.39	0.41	0.16	2.62	1.65
F6	0.17	0.11	0.27	0.23	1.59	0.17	0.39	0.42	0.37	0.20	2.90	1.94
F7	0.13	0.06	0.16	0.17	0.91	0.14	0.31	0.45	0.26	0.20	3.93	1.52
F8	0.14	0.07	0.16	0.19	0.90	0.10	0.29	0.36	0.19	0.17	4.81	1.41
F9	0.12	0.06	0.17	0.18	0.93	0.10	0.29	0.33	0.26	0.19	4.39	1.51
F10	0.13	0.08	0.19	0.19	1.08	0.11	0.34	0.36	0.22	0.18	4.74	1.92
F11	0.13	0.06	0.14	0.19	0.82	0.09	0.30	0.26	0.24	0.17	2.83	1.41
F12	0.13	0.06	0.16	0.22	0.94	0.11	0.34	0.19	0.17	0.15	2.33	1.58
F13	0.12	0.06	0.15	0.19	0.77	0.09	0.30	0.15	0.13	0.15	2.76	1.56
F14	0.14	0.08	0.20	0.21	1.07	0.12	0.31	0.25	0.14	0.18	2.57	1.70
F15	0.11	0.05	0.15	0.19	0.74	0.07	0.27	0.23	0.14	0.16	2.70	1.39
F16	0.13	0.07	0.17	0.19	0.86	0.09	0.29	0.43	0.16	0.17	4.37	1.47
F17	0.18	0.10	0.25	0.23	1.57	0.15	0.36	0.34	0.21	0.16	3.40	1.64
F18	0.12	0.06	0.16	0.21	0.87	0.09	0.28	0.26	0.19	0.15	2.48	1.69
F19	0.36	0.29	0.78	0.47	4.61	0.50	0.77	0.38	0.30	0.32	2.53	2.60
F20	0.17	0.10	0.29	0.22	1.53	0.17	0.38	0.49	0.24	0.20	4.53	1.68
F21	0.20	0.13	0.31	0.24	1.82	0.21	0.46	0.43	0.33	0.21	3.83	1.63
F22	0.21	0.14	0.36	0.31	2.13	0.23	0.47	0.45	0.35	0.22	3.43	1.99
F23	0.23	0.16	0.38	0.28	2.36	0.26	0.49	0.42	0.38	0.22	3.05	1.96
F24	0.24	0.18	0.43	0.31	2.64	0.28	0.53	0.42	0.31	0.23	2.59	2.00
F25	0.23	0.14	0.34	0.27	2.17	0.25	0.50	0.32	0.31	0.21	2.65	1.95
F26	0.25	0.19	0.46	0.31	2.94	0.33	0.57	0.46	0.39	0.22	3.31	1.98
F27	0.27	0.21	0.49	0.34	0.68	0.34	0.60	0.45	0.44	0.24	1.83	2.28
F28	0.25	0.18	0.41	0.31	2.80	0.30	0.54	0.33	0.39	0.22	1.85	1.79
F29	0.22	0.15	0.36	0.27	2.31	0.24	0.49	0.43	0.29	0.25	2.72	1.97
F30	0.41	0.35	0.88	0.48	5.42	0.59	0.85	0.44	0.63	0.34	2.57	2.97
Mean	0.18	0.11	0.28	0.24	1.60	0.19	0.40	0.35	0.28	0.21	3.08	1.80

Table 5
Comparison of results on CEC2017 ($D = 30$).

ID	index	GWO	WOA	HHO	IPO	MSMA	COA	OMA	LSHADE _cnEpSin	LSHADE _SPACMA	MadDE	EA4eig	BHO
CEC2017-F1	Best	1.3589E+08	3.3702E+09	1.3566E+08	1.8913E+03	3.3287E+02	1.9454E+07	8.6365E+08	1.6058E+02	**1.0285E+02**	1.9514E+05	1.1481E+04	1.9899E+02
	Worst	7.5419E+09	9.4280E+09	1.3269E+09	5.9805E+06	2.1515E+04	1.0683E+09	2.9825E+09	1.9487E+04	6.4958E+03	2.6445E+06	1.1418E+05	**4.6304E+03**
	Ave	3.2260E+09	6.0897E+09	5.2627E+08	6.5631E+05	8.0335E+03	3.5549E+08	1.7716E+09	5.7434E+03	**1.5608E+03**	6.8887E+05	4.2660E+04	1.9782E+03
	Std	2.5925E+09	1.9345E+09	3.9721E+08	1.8729E+06	8.7834E+03	3.5709E+08	6.6450E+08	7.2671E+03	2.0740E+03	7.0580E+05	2.9585E+04	**1.3788E+03**
CEC2017-F2	Best	2.3099E+25	4.9424E+28	2.5364E+24	3.5764E+19	3.6612E+10	1.3316E+19	2.2301E+22	2.2312E+12	6.1071E+44	1.0624E+17	3.5856E+12	**1.3560E+07**
	Worst	1.4419E+32	8.6095E+36	1.4446E+33	1.5091E+32	**4.3409E+15**	2.3856E+26	2.0504E+26	2.7459E+19	1.6116E+54	5.0097E+20	1.8978E+17	1.6739E+18
	Ave	1.4594E+31	8.7752E+35	1.9473E+32	1.5177E+31	**4.4069E+14**	2.5636E+25	4.9086E+25	5.2973E+18	1.6223E+53	5.1443E+19	3.5830E+16	1.6763E+17
	Std	4.5481E+31	2.7146E+36	4.6264E+32	4.7651E+31	**1.3719E+15**	7.4780E+25	7.8958E+25	9.5328E+18	5.1044E+53	1.5807E+20	6.2973E+16	5.2962E+17
CEC2017-F3	Best	5.0249E+04	1.6963E+05	5.0388E+04	2.9652E+04	2.4775E+04	7.8309E+04	5.4109E+04	1.0296E+03	**3.1036E+02**	7.9551E+04	8.1183E+03	3.8559E+03
	Worst	7.3979E+04	4.4839E+05	6.8726E+04	4.0612E+04	5.3542E+04	1.7662E+05	7.9515E+04	6.0120E+04	1.1411E+05	1.0341E+05	2.7681E+04	**1.7482E+04**
	Ave	6.3760E+04	2.9108E+05	5.6555E+04	3.4080E+04	3.7472E+04	1.2841E+05	6.9115E+04	1.6185E+04	2.2903E+04	9.1373E+04	1.5618E+04	**8.8063E+03**
	Std	9.9689E+03	8.2797E+04	7.0082E+03	**3.4220E+03**	1.0207E+04	3.8222E+04	8.3107E+03	2.2890E+04	4.7372E+04	9.1690E+03	7.1706E+03	4.1290E+03
CEC2017-F4	Best	5.2567E+02	7.8702E+02	6.3031E+02	4.8240E+02	4.7067E+02	5.3231E+02	6.5289E+02	4.6852E+02	**4.0776E+02**	4.7386E+02	4.5574E+02	4.6928E+02
	Worst	8.8773E+02	1.6736E+03	7.6796E+02	5.4081E+02	5.9964E+02	7.5531E+02	9.6295E+02	5.4381E+02	5.4788E+02	5.3656E+02	**5.1739E+02**	5.1905E+02
	Ave	6.4661E+02	1.1517E+03	6.8146E+02	5.1103E+02	5.0538E+02	6.0780E+02	8.0121E+02	4.9443E+02	5.0923E+02	5.0670E+02	**4.9034E+02**	4.9287E+02
	Std	1.2170E+02	2.7516E+02	5.2756E+01	2.1641E+01	3.8338E+01	7.0293E+01	1.1055E+02	2.3886E+01	3.7753E+01	1.9180E+01	2.1513E+01	**1.7914E+01**
CEC2017-F5	Best	5.9220E+02	7.7647E+02	7.3757E+02	7.2393E+02	5.6278E+02	6.7103E+02	6.7370E+02	5.3323E+02	5.3088E+02	6.5821E+02	6.0538E+02	**5.2434E+02**
	Worst	7.5208E+02	9.2779E+02	8.1659E+02	8.7620E+02	7.3956E+02	7.9977E+02	7.4965E+02	**5.6816E+02**	5.8452E+02	6.9590E+02	6.6523E+02	6.4051E+02
	Ave	6.4763E+02	8.3983E+02	7.8521E+02	7.7020E+02	6.7694E+02	7.3956E+02	7.1219E+02	**5.4951E+02**	5.5962E+02	6.8156E+02	6.3447E+02	5.6496E+02
	Std	4.6090E+01	5.2061E+01	2.7410E+01	4.3257E+01	5.3908E+01	4.5880E+01	2.5154E+01	**1.2414E+01**	1.5043E+01	1.2768E+01	1.7818E+01	3.0115E+01
CEC2017-F6	Best	6.0843E+02	6.6507E+02	6.5744E+02	6.4660E+02	6.1263E+02	6.2630E+02	6.1996E+02	6.0095E+02	**6.0059E+02**	6.0708E+02	6.0104E+02	6.0397E+02
	Worst	6.2177E+02	7.0554E+02	6.7237E+02	6.6707E+02	6.3992E+02	6.6417E+02	6.3948E+02	6.0700E+02	**6.0279E+02**	6.2056E+02	6.0607E+02	6.0964E+02
	Ave	6.1414E+02	6.8196E+02	6.6767E+02	6.5795E+02	6.2866E+02	6.5378E+02	6.2755E+02	6.0384E+02	**6.0125E+02**	6.1225E+02	6.0272E+02	6.0625E+02
	Std	3.5246E+00	1.0268E+01	4.7099E+00	7.4018E+00	7.3831E+00	1.2350E+01	7.1063E+00	2.2537E+00	**7.1107E-01**	4.2579E+00	1.3565E+00	2.4323E+00
CEC2017-F7	Best	8.1277E+02	1.2285E+03	1.2511E+03	1.0832E+03	9.2904E+02	1.0798E+03	1.1125E+03	7.8435E+02	**7.7468E+02**	9.3840E+02	8.5285E+02	7.8898E+02
	Worst	9.6326E+02	1.5512E+03	1.4560E+03	1.1570E+03	1.1817E+03	1.3425E+03	1.4384E+03	**8.3378E+02**	8.4799E+02	1.0306E+03	9.3394E+02	8.3479E+02
	Ave	8.8761E+02	1.3180E+03	1.3455E+03	1.1106E+03	1.0538E+03	1.2497E+03	1.1914E+03	8.1136E+02	**8.0540E+02**	9.6776E+02	8.9234E+02	8.1615E+02
	Std	4.3414E+01	9.7541E+01	6.1784E+01	2.4065E+01	9.8575E+01	9.3584E+01	9.7034E+01	1.8290E+01	2.6252E+01	3.3239E+01	2.2129E+01	**1.6169E+01**
CEC2017-F8	Best	8.6493E+02	9.8697E+02	9.4336E+02	9.5077E+02	8.9919E+02	9.1645E+02	9.5984E+02	**8.2654E+02**	8.2723E+02	9.1659E+02	8.6242E+02	8.3263E+02
	Worst	9.9452E+02	1.0854E+03	1.0375E+03	1.0072E+03	9.6529E+02	1.0027E+03	1.0133E+03	**8.6732E+02**	8.9007E+02	9.5802E+02	9.7247E+02	9.2278E+02
	Ave	9.0072E+02	1.0306E+03	9.8331E+02	9.7465E+02	9.3823E+02	9.7556E+02	9.8712E+02	**8.4779E+02**	8.5875E+02	9.4379E+02	9.2341E+02	8.7160E+02
	Std	3.6356E+01	3.3400E+01	2.7452E+01	1.6730E+01	2.4806E+01	2.8731E+01	1.6304E+01	1.4817E+01	1.7238E+01	**1.4772E+01**	3.2285E+01	3.2290E+01
CEC2017-F9	Best	1.2389E+03	5.7650E+03	7.7007E+03	4.2573E+03	2.7210E+03	4.0443E+03	2.3722E+03	1.0519E+03	1.0559E+03	2.5718E+03	**9.1532E+02**	1.0681E+03
	Worst	3.3251E+03	1.8988E+04	1.1724E+04	5.8467E+03	5.2073E+03	9.7577E+03	6.0441E+03	2.5084E+03	1.6883E+03	4.6172E+03	**1.2125E+03**	2.2442E+03
	Ave	2.2312E+03	1.1471E+04	8.4663E+03	4.9711E+03	3.8263E+03	7.5561E+03	4.4647E+03	1.3674E+03	1.3053E+03	3.5023E+03	**1.0147E+03**	1.4096E+03
	Std	7.9882E+02	4.3665E+03	1.2073E+03	4.3754E+02	7.7360E+02	1.6205E+03	1.2627E+03	4.2483E+02	2.3667E+02	7.2188E+02	**8.9908E+01**	4.0521E+02
CEC2017-F10	Best	3.7082E+03	6.7696E+03	5.2767E+03	4.4187E+03	4.3598E+03	5.1939E+03	7.4403E+03	3.2605E+03	3.3671E+03	5.1255E+03	5.1061E+03	**3.1358E+03**
	Worst	8.5193E+03	9.4784E+03	6.7740E+03	6.5670E+03	6.2378E+03	6.7092E+03	9.1402E+03	**4.7591E+03**	8.8080E+03	6.2626E+03	7.2281E+03	5.6288E+03
	Ave	4.9112E+03	7.7048E+03	5.9434E+03	5.4700E+03	5.2236E+03	5.9388E+03	8.5501E+03	**3.8547E+03**	5.2347E+03	5.7174E+03	6.3965E+03	4.5025E+03
	Std	1.3464E+03	8.2001E+02	5.7421E+02	6.4379E+02	5.5255E+02	5.7958E+02	5.2712E+02	5.2209E+02	1.5473E+03	**4.5244E+02**	5.9110E+02	6.9381E+02
CEC2017-F11	Best	1.4538E+03	5.5313E+03	1.4384E+03	1.1761E+03	1.2122E+03	1.3395E+03	1.3011E+03	1.1675E+03	1.1678E+03	1.2626E+03	**1.1577E+03**	1.1888E+03
	Worst	3.8655E+03	1.3188E+04	1.7883E+03	1.2971E+03	1.3142E+03	2.1875E+03	1.5459E+03	1.4214E+03	1.4721E+03	1.4099E+03	1.3040E+03	**1.2520E+03**
	Ave	2.3192E+03	8.6903E+03	1.6073E+03	**1.2116E+03**	1.2689E+03	1.6204E+03	1.4192E+03	1.2859E+03	1.2723E+03	1.3272E+03	1.2151E+03	1.2148E+03
	Std	9.0477E+02	2.5854E+03	1.1787E+02	4.2336E+01	3.8505E+01	2.5465E+02	8.1443E+01	7.8489E+01	9.6729E+01	4.9361E+01	4.7875E+01	**2.7644E+01**
CEC2017-F12	Best	2.0435E+07	1.6966E+08	4.0132E+07	2.1875E+05	3.6708E+05	2.1939E+06	2.9852E+06	**1.4064E+04**	2.6905E+04	8.8511E+05	6.8529E+04	2.7151E+04
	Worst	1.6500E+09	9.9424E+08	5.6579E+08	5.0572E+06	6.7990E+06	3.6188E+07	3.8904E+07	2.0594E+06	8.5818E+05	3.0315E+06	2.1578E+06	**8.0318E+05**
	Ave	2.6957E+08	4.5709E+08	1.8236E+08	1.2479E+06	3.7890E+06	1.1635E+07	1.2710E+07	3.9823E+05	**1.6515E+05**	2.0708E+06	4.4301E+05	2.1362E+05
	Std	4.9791E+08	2.3511E+08	1.5948E+08	1.4480E+06	2.6267E+06	1.2689E+07	1.0894E+07	5.9975E+05	**2.5593E+05**	5.7779E+05	6.0976E+05	2.5880E+05
CEC2017-F13	Best	9.8254E+04	6.8231E+05	4.0271E+05	9.7433E+03	7.6858E+03	3.9962E+04	2.6797E+04	3.8579E+03	**2.4802E+03**	2.4546E+04	6.7075E+03	6.6747E+03
	Worst	1.2503E+08	7.4343E+07	6.9189E+06	**2.1665E+04**	6.6086E+04	7.5560E+06	1.2906E+07	7.0995E+04	5.6533E+04	1.7885E+05	2.5046E+04	4.8234E+04
	Ave	2.2517E+07	1.6499E+07	1.5158E+06	1.5373E+04	2.8612E+04	8.3246E+05	1.3476E+06	1.7491E+04	2.2929E+04	6.4329E+04	**1.3486E+04**	1.7758E+04
	Std	4.7635E+07	2.1307E+07	1.9151E+06	**3.7120E+03**	2.5010E+04	2.3698E+06	4.0552E+06	2.0003E+04	1.7069E+04	4.7803E+04	5.6552E+03	1.2770E+04
CEC2017-F14	Best	1.4816E+04	8.5824E+04	3.2314E+04	4.8172E+03	1.0316E+04	8.3947E+04	3.5272E+03	1.5555E+03	1.5249E+03	9.2593E+03	**1.4973E+03**	1.5023E+03

(*continued on next page*)

Table 5 (*continued*)

ID	index	GWO	WOA	HHO	IPO	MSMA	COA	OMA	LSHADE _cnEpSin	LSHADE _SPACMA	MadDE	EA4eig	BHO
	Worst	1.3626E+06	7.3659E+06	3.5478E+06	2.7835E+04	9.2893E+05	6.3549E+05	3.4138E+04	1.7995E+03	1.7206E+03	3.4931E+05	**1.6058E+03**	1.7284E+03
	Ave	3.2967E+05	2.1249E+06	1.4473E+06	1.4926E+04	2.4949E+05	2.1598E+05	1.3184E+04	1.6145E+03	1.5882E+03	1.5991E+05	**1.5388E+03**	1.6125E+03
	Std	4.4240E+05	2.2037E+06	1.0450E+06	8.2441E+03	3.5396E+05	1.5934E+05	1.2336E+04	6.8070E+01	6.6197E+01	1.4038E+05	**3.1198E+01**	7.1848E+01
CEC2017-F15	Best	1.6929E+04	7.5165E+05	9.7411E+04	1.8771E+03	2.1797E+03	7.4927E+03	2.9499E+03	**1.6481E+03**	1.7285E+03	2.4946E+03	1.7500E+03	1.7798E+03
	Worst	8.3518E+06	4.4166E+07	2.6764E+05	3.6622E+03	2.3736E+04	7.1489E+04	2.6453E+04	3.3042E+03	**2.2673E+03**	1.8256E+04	6.2302E+03	3.6977E+03
	Ave	2.0859E+06	9.9611E+06	1.7314E+05	2.6650E+03	9.6822E+03	2.8255E+04	8.5084E+03	2.3436E+03	**1.9623E+03**	9.9305E+03	2.4304E+03	2.5680E+03
	Std	2.8104E+06	1.3891E+07	6.3310E+04	6.7241E+02	8.3065E+03	2.3818E+04	7.1690E+03	4.9712E+02	**1.9558E+02**	4.5708E+03	1.3535E+03	5.9530E+02
CEC2017-F16	Best	2.1319E+03	3.1705E+03	3.5485E+03	2.6917E+03	2.2446E+03	2.1942E+03	2.8157E+03	**1.9311E+03**	2.0724E+03	2.2838E+03	2.2492E+03	1.9585E+03
	Worst	3.9107E+03	4.7486E+03	4.3896E+03	3.8626E+03	3.1975E+03	3.5287E+03	3.6313E+03	2.6138E+03	3.6786E+03	2.9225E+03	2.9138E+03	2.5032E+03
	Ave	2.7588E+03	4.0013E+03	3.8658E+03	3.2094E+03	2.7155E+03	2.8282E+03	3.3513E+03	2.2944E+03	2.8792E+03	2.6890E+03	2.6431E+03	2.2852E+03
	Std	5.1703E+02	5.8549E+02	3.0189E+02	3.3467E+02	2.8250E+02	5.0966E+02	2.7645E+02	2.2619E+02	5.3386E+02	2.0000E+02	1.8758E+02	1.8000E+02
CEC2017-F17	Best	1.8728E+03	2.0918E+03	2.3508E+03	2.2946E+03	1.9679E+03	2.0014E+03	1.8705E+03	1.7848E+03	1.8324E+03	1.8665E+03	1.8736E+03	**1.7701E+03**
	Worst	2.5049E+03	3.1163E+03	2.9869E+03	3.5219E+03	2.6539E+03	2.4227E+03	2.4622E+03	2.0663E+03	2.9535E+03	2.2017E+03	2.2664E+03	**1.9794E+03**
	Ave	2.2012E+03	2.7714E+03	2.6754E+03	2.7712E+03	2.2838E+03	2.1928E+03	2.1821E+03	1.9487E+03	2.4981E+03	2.0860E+03	2.0649E+03	**1.8600E+03**
	Std	2.1111E+02	2.8841E+02	2.2982E+02	3.5274E+02	2.4144E+02	1.5683E+02	1.8599E+02	8.2948E+01	3.8688E+02	1.2601E+02	1.4002E+02	**5.6618E+01**
CEC2017-F18	Best	1.4285E+05	4.6152E+05	2.1449E+05	8.1473E+04	3.8721E+05	1.4546E+05	3.7887E+04	4.7884E+03	**3.9434E+03**	6.1470E+04	1.1345E+04	5.2845E+03
	Worst	6.8751E+06	5.5835E+07	8.9458E+06	3.8020E+05	1.1837E+07	5.4586E+06	6.5113E+05	4.4235E+04	**3.0757E+04**	1.5422E+06	6.2284E+04	7.5240E+04
	Ave	2.1586E+06	1.4722E+07	3.2075E+06	1.6464E+05	4.3876E+06	2.3253E+06	2.8110E+05	1.9897E+04	**1.3177E+04**	6.1518E+05	3.2047E+04	4.2682E+04
	Std	2.6335E+06	1.7562E+07	2.8126E+06	1.0646E+05	4.2308E+06	1.9247E+06	1.9836E+05	1.3237E+04	**8.1834E+03**	5.2767E+05	1.9148E+04	2.3125E+04
CEC2017-F19	Best	4.6140E+04	3.5189E+06	1.2306E+06	3.0203E+03	2.0397E+03	1.1887E+04	2.3865E+03	**1.9695E+03**	2.0195E+03	5.1365E+03	2.0044E+03	1.9943E+03
	Worst	3.2221E+06	1.0473E+08	4.4967E+06	1.0376E+04	5.6503E+04	6.9506E+04	4.7342E+04	**2.4153E+03**	3.0282E+03	1.8836E+04	9.7342E+03	2.7969E+03
	Ave	1.1112E+06	3.3851E+07	2.6589E+06	6.6711E+03	1.3677E+04	2.4844E+04	1.6657E+04	2.1901E+03	**2.1765E+03**	1.0111E+04	3.1900E+03	2.2562E+03
	Std	1.0740E+06	3.0198E+07	1.0275E+06	2.4700E+03	1.6222E+04	1.8118E+04	1.3275E+04	**1.4404E+02**	3.0390E+02	5.1609E+03	2.3782E+03	2.3024E+02
CEC2017-F20	Best	2.2599E+03	2.6082E+03	2.4995E+03	2.7346E+03	2.2364E+03	2.2132E+03	2.4576E+03	2.1009E+03	2.4355E+03	2.2491E+03	2.2387E+03	**2.0908E+03**
	Worst	2.8844E+03	3.3750E+03	3.0801E+03	3.2675E+03	2.8562E+03	2.9633E+03	2.8915E+03	**2.3760E+03**	3.3209E+03	2.5446E+03	2.4389E+03	2.3862E+03
	Ave	2.4739E+03	2.9143E+03	2.7984E+03	2.9916E+03	2.4818E+03	2.7655E+03	2.6730E+03	**2.2102E+03**	2.9284E+03	2.3627E+03	2.3513E+03	2.2550E+03
	Std	1.9631E+02	2.3711E+02	2.1982E+02	1.6611E+02	1.5579E+02	2.3333E+02	1.3111E+02	9.6641E+01	2.8571E+02	1.0254E+02	**7.1496E+01**	1.1084E+02
CEC2017-F21	Best	2.3673E+03	2.5240E+03	2.5356E+03	2.5401E+03	2.4111E+03	2.3656E+03	2.4264E+03	**2.3307E+03**	2.3362E+03	2.4427E+03	2.4184E+03	2.3555E+03
	Worst	2.4524E+03	2.6754E+03	2.6606E+03	2.6311E+03	2.5004E+03	2.6397E+03	2.5262E+03	**2.3737E+03**	2.4048E+03	2.4887E+03	2.4606E+03	2.4063E+03
	Ave	2.4101E+03	2.6166E+03	2.6030E+03	2.5833E+03	2.4518E+03	2.4848E+03	2.4805E+03	**2.3544E+03**	2.3613E+03	2.4649E+03	2.4425E+03	2.3704E+03
	Std	2.6499E+01	4.7360E+01	4.4941E+01	2.8922E+01	3.2100E+01	7.7810E+01	3.1429E+01	**1.2500E+01**	2.0743E+01	1.5725E+01	1.4576E+01	1.4007E+01
CEC2017-F22	Best	2.8050E+03	6.1203E+03	7.0412E+03	6.5387E+03	2.3157E+03	2.4401E+03	2.5593E+03	2.3005E+03	**2.3079E+03**	2.3192E+03	2.3054E+03	2.3009E+03
	Worst	6.5223E+03	9.7256E+03	9.0811E+03	8.3451E+03	7.0131E+03	8.9133E+03	3.1206E+03	5.4528E+03	6.3712E+03	2.3429E+03	2.3124E+03	**2.3179E+03**
	Ave	4.4380E+03	8.7048E+03	8.1406E+03	7.4520E+03	4.7732E+03	4.4805E+03	2.8289E+03	3.4089E+03	2.9373E+03	2.3202E+03	2.3006E+03	**2.3052E+03**
	Std	1.6801E+03	9.8679E+02	7.5125E+02	4.9997E+02	2.1559E+03	2.6201E+03	2.3204E+02	1.4242E+03	1.4099E+03	8.5652E+00	**2.2890E+00**	2.3827E+00
CEC2017-F23	Best	2.7467E+03	2.9353E+03	3.0964E+03	3.4153E+03	2.7361E+03	2.7468E+03	2.8356E+03	2.7025E+03	2.7268E+03	2.7967E+03	2.7608E+03	**2.7093E+03**
	Worst	2.8264E+03	3.4416E+03	3.6854E+03	3.8316E+03	2.9030E+03	2.9695E+03	2.9976E+03	**2.7873E+03**	2.7946E+03	2.8556E+03	2.8120E+03	2.8071E+03
	Ave	2.7852E+03	3.1574E+03	3.3214E+03	3.6191E+03	2.8054E+03	2.8695E+03	2.9473E+03	**2.7346E+03**	2.7495E+03	2.8272E+03	2.7881E+03	2.7454E+03
	Std	2.9271E+01	1.5002E+02	1.9688E+02	1.4523E+02	5.5998E+01	6.9153E+01	4.8451E+01	2.2891E+01	2.5297E+01	**2.0561E+01**	2.1580E+01	3.3072E+01
CEC2017-F24	Best	2.9061E+03	3.1312E+03	3.2522E+03	3.1606E+03	2.9048E+03	2.9044E+03	3.0309E+03	2.8675E+03	2.8746E+03	2.9458E+03	2.8735E+03	**2.8504E+03**
	Worst	3.0836E+03	3.4045E+03	3.8256E+03	3.4549E+03	3.0413E+03	3.0908E+03	3.2325E+03	**2.9248E+03**	3.2081E+03	3.0394E+03	2.9856E+03	2.9361E+03
	Ave	2.9770E+03	3.2548E+03	3.5442E+03	3.3446E+03	2.9699E+03	3.0104E+03	3.1091E+03	**2.8931E+03**	2.9639E+03	2.9926E+03	2.9364E+03	2.9046E+03
	Std	7.3497E+01	9.2559E+01	1.7690E+02	8.8913E+01	5.0176E+01	5.9583E+01	5.4013E+01	**2.1166E+01**	9.7446E+01	2.6595E+01	3.5917E+01	2.3583E+01
CEC2017-F25	Best	2.9644E+03	3.1469E+03	2.9611E+03	2.8917E+03	2.8939E+03	2.9268E+03	2.9593E+03	2.8838E+03	2.8899E+03	2.8991E+03	2.8926E+03	**2.8848E+03**
	Worst	3.3543E+03	3.3326E+03	3.1616E+03	2.9583E+03	2.9617E+03	3.0389E+03	3.1221E+03	2.9599E+03	2.9361E+03	2.9512E+03	2.9180E+03	**2.9072E+03**
	Ave	3.0555E+03	3.2326E+03	3.0250E+03	2.9362E+03	2.9268E+03	2.9812E+03	3.0633E+03	2.9012E+03	2.9006E+03	2.9234E+03	2.8990E+03	**2.8972E+03**
	Std	1.1187E+02	6.9560E+01	6.1697E+01	2.0558E+01	2.3137E+01	3.2885E+01	6.2614E+01	2.3106E+01	1.7559E+01	2.0336E+01	7.8788E+00	**2.9620E+00**
CEC2017-F26	Best	4.1863E+03	6.6122E+03	3.6470E+03	3.4253E+03	3.2383E+03	4.1656E+03	5.4106E+03	3.9114E+03	2.9076E+03	2.8304E+03	4.6805E+03	**2.8034E+03**
	Worst	5.9685E+03	1.0168E+04	1.0389E+04	9.2924E+03	6.9759E+03	7.0785E+03	7.5705E+03	**4.7785E+03**	5.1252E+03	5.7223E+03	5.5593E+03	5.1832E+03
	Ave	5.0539E+03	8.5337E+03	7.7298E+03	6.6543E+03	5.0853E+03	5.4324E+03	6.4472E+03	4.3832E+03	4.5295E+03	**3.3815E+03**	4.9886E+03	3.8029E+03
	Std	5.0985E+02	1.1760E+03	1.8675E+03	2.3057E+03	1.2063E+03	1.0449E+03	6.2045E+02	**2.6749E+02**	6.3406E+02	8.6821E+02	2.7677E+02	9.6583E+02
CEC2017-F27	Best	3.2465E+03	3.3563E+03	3.3713E+03	3.4144E+03	3.2274E+03	3.2266E+03	3.2404E+03	3.2214E+03	3.2150E+03	3.2402E+03	**3.1602E+03**	3.2145E+03
	Worst	3.3323E+03	3.7790E+03	4.1610E+03	5.1510E+03	3.2546E+03	3.2973E+03	3.4562E+03	3.2580E+03	4.3283E+03	3.2858E+03	**3.2064E+03**	3.2542E+03
	Ave	3.2842E+03	3.5144E+03	3.6293E+03	4.3423E+03	3.2393E+03	3.2755E+03	3.3289E+03	3.2369E+03	3.4721E+03	3.2509E+03	**3.2097E+03**	3.2391E+03

(*continued on next page*)

Table 5 (*continued*)

ID	index	GWO	WOA	HHO	IPO	MSMA	COA	OMA	LSHADE_cnEpSin	LSHADE_SPACMA	MadDE	EA4eig	BHO
	Std	3.0781E+01	1.2947E+02	2.1923E+02	5.3369E+02	1.1987E+01	2.3515E+01	5.8085E+01	1.2021E+01	4.0432E+02	1.1618E+01	**1.1352E+01**	1.4280E+01
CEC2017-F28	Best	3.3321E+03	3.7037E+03	3.3846E+03	3.2994E+03	3.2290E+03	3.2764E+03	3.4127E+03	3.2069E+03	**3.1160E+03**	3.2289E+03	3.2228E+03	3.2037E+03
	Worst	4.0924E+03	4.2640E+03	3.7508E+03	3.3427E+03	3.3236E+03	3.4987E+03	4.0429E+03	3.2568E+03	1.1022E+04	3.3015E+03	3.2948E+03	**3.2561E+03**
	Ave	3.5539E+03	3.9917E+03	3.5458E+03	3.3177E+03	3.2822E+03	3.3634E+03	3.6066E+03	3.2217E+03	5.2284E+03	3.2841E+03	3.2408E+03	**3.2295E+03**
	Std	2.3141E+02	1.8241E+02	1.1634E+02	1.8171E+01	4.0176E+01	6.7637E+01	1.8710E+02	**1.4734E+01**	3.2376E+03	2.2656E+01	2.4308E+01	1.5525E+01
CEC2017-F29	Best	3.6499E+03	4.5736E+03	4.0801E+03	3.9564E+03	3.7156E+03	3.9174E+03	3.9128E+03	**3.4118E+03**	3.5571E+03	3.6725E+03	3.6785E+03	3.5598E+03
	Worst	4.6441E+03	5.9117E+03	5.7263E+03	5.0420E+03	4.6003E+03	4.7469E+03	4.4696E+03	4.1167E+03	4.5298E+03	3.9894E+03	3.9411E+03	**3.9444E+03**
	Ave	3.9224E+03	5.3129E+03	4.9381E+03	4.6381E+03	4.1442E+03	4.2777E+03	4.2072E+03	**3.6880E+03**	4.0704E+03	3.8364E+03	3.7978E+03	3.7330E+03
	Std	2.8452E+02	4.8527E+02	4.7395E+02	3.3416E+02	3.0478E+02	2.7913E+02	2.2058E+02	2.1494E+02	3.0836E+02	1.1002E+02	**7.7007E+01**	1.2371E+02
CEC2017-F30	Best	1.6967E+06	2.0412E+07	3.1079E+06	7.7985E+03	1.0327E+04	1.3054E+05	3.9249E+04	8.2377E+03	**7.6725E+03**	8.8083E+04	1.3493E+04	1.0333E+04
	Worst	3.5921E+07	1.7105E+08	4.8324E+07	3.3371E+04	7.4418E+04	1.7702E+06	2.5703E+05	2.7981E+04	4.0065E+04	4.3505E+05	6.4363E+04	**2.2780E+04**
	Ave	1.2823E+07	7.1165E+07	1.3947E+07	1.5495E+04	3.7015E+04	9.1280E+05	1.1250E+05	1.8946E+04	1.8540E+04	2.7837E+05	2.6427E+04	**1.4482E+04**
	Std	1.1645E+07	4.5546E+07	1.3781E+07	8.4260E+03	2.2838E+04	5.6380E+05	6.7275E+04	6.7098E+03	9.2854E+03	1.0158E+05	1.5072E+04	**4.3924E+03**
average Best rank		7.400	11.400	10.500	8.067	5.800	7.833	8.600	**2.267**	2.600	6.700	4.367	2.467
average Worst rank		8.300	11.533	10.033	7.267	6.133	8.267	8.000	2.700	5.200	5.167	3.267	**2.133**
average Ave rank		7.400	11.567	10.233	7.633	6.067	8.300	8.300	2.533	4.367	5.933	3.300	**2.367**
average Std rank		9.033	10.933	8.800	6.367	7.167	8.600	6.900	3.267	6.200	4.267	3.233	**3.233**

group selection pressure, such structural advantages are difficult to translate into sustained global improvement, which explains the limited gains of partial configurations. When Explorpolis is enabled alone, its acceptance rule emphasizes de-redundancy; without QPRS to provide effective inter-group coordination, this may manifest as overly restrictive filtering and reduced effective updates, leading to a degraded overall ranking. Similarly, QPRS mainly contributes at the inter-group level by promoting stable coordination, but it relies on informative structure and sufficient diversity as inputs; therefore, enabling QPRS alone does not produce a pronounced improvement. In $D = 100$, the above coupling becomes amplified: K-means initialization is more likely to introduce distribution bias or reduce diversity in high-dimensional spaces, and without Explorpolis for correction and QPRS for stabilized coordination, the degradation becomes more evident. Only the full BHO forms a functional closed loop. In this synergy, K-means provides a structured start, Explorpolis prevents homogeneity, and QPRS ensures reliable selection pressure. This integration explains why the complete algorithm consistently achieves optimal performance across both dimensions.

4.3. Assessing performance with CEC 2017 test suite

This section performs a performance evaluation of the BHO algorithm using the CEC2017 test set and compares it with various advanced MH algorithms through statistical analysis. Specifically, performance evaluation tables for multiple MH algorithms were constructed based on statistical indicators such as the optimal value, worst value, standard deviation, and mean of the optimization results. The ranking results for $D = 30$ and $D = 100$ scenarios are shown in Tables 5 and 6, respectively, with the best values for each test function highlighted in bold.

In the 30-dimensional CEC2017 test, BHO achieved the best performance in 7 functions (23.33%), ranked second in 10 functions (33.33%), and ranked third in 8 functions (26.67%), resulting in the coverage of the top three reaches 83.33%.. Its Friedman mean rank was 2.367, the lowest among all competitors. In comparison, LSHADE_cnEpSin obtained a mean rank of 2.533 with a top-three coverage of 80%. Thus, BHO exceeded LSHADE_cnEpSin by 6.55% and consistently outperformed the remaining algorithms, confirming its superior global search capability on complex optimization tasks.

To further assess its performance in high-dimensional scenarios, experiments were conducted on the 100-dimensional CEC2017 test functions under the same settings (results in Table 6). BHO attained the best results on 13 functions (43.33%), ranked second on 8 functions (26.67%), and ranked third on the remainder, yielding a 100% top-three coverage. Its Friedman mean rank was 1.867, the lowest among all algorithms. In contrast, LSHADE_cnEpSin ranked second with a mean rank of 2.000 and a top-three coverage of 83.33%. Hence, BHO surpassed LSHADE_cnEpSin by 0.133 in average rank and by 16.67% in top-three coverage, confirming its sustained global search capability and strong generalization in high-dimensional optimization. The convergence curves (Fig. 6) further demonstrate that BHO achieves faster early-stage descent and more stable late-stage convergence in most functions; while the Friedman average ranking radar chart (Fig. 7) shows that BHO encloses the smallest area and achieves the best overall ranking, reaffirming its superiority in high-dimensional complex optimization.

To further analyze the convergence behavior of the BHO within 500D evaluations, the average optimization descent rate of each algorithm across 30 functions was computed as $\mu^k = \overline{\frac{F^k - F^{k-1}}{F^k - 1}}$. The results are presented in Fig. 8. In terms of overall trends, all algorithms exhibited a three-phase convergence pattern: a rapid decline during 1–100D, steady convergence during 101–400D, and slight fluctuations during 401–500D BHO demonstrated clear advantages across all three phases. In the first stage, its descent rate consistently ranked among the top two, and at 30D its Friedman average ranking surpassed LSHADE_SPACMA, second only to LSHADE_cnEpSin; In the second stage, BHO remained steadily within

Table 6
Comparison of results on CEC2017 ($D = 100$).

ID	index	GWO	WOA	HHO	IPO	MSMA	COA	OMA	LSHADE _cnEpSin	LSHADE _SPACMA	MadDE	EA4eig	BHO
CEC2017-F1	Best	3.0672E+10	2.9372E+10	7.0796E+09	5.1808E+06	2.5034E+06	2.1920E+10	5.2134E+10	2.1946E+02	2.0211E+02	6.3715E+09	1.9902E+08	**2.0038E+02**
	Worst	4.4129E+10	5.7954E+10	1.1877E+10	1.8511E+07	6.9675E+08	3.1270E+10	6.7136E+10	**2.6815E+04**	4.5621E+04	8.0851E+09	4.9054E+08	2.7711E+04
	Ave	3.4453E+10	4.2686E+10	9.7232E+09	1.1816E+07	3.8539E+08	2.5664E+10	5.9856E+10	8.9460E+03	1.3039E+04	7.2537E+09	2.9312E+08	**7.1656E+03**
	Std	5.7598E+09	1.3133E+10	1.9804E+09	6.0722E+06	2.9396E+08	3.5626E+09	6.1443E+09	**1.0748E+04**	1.9352E+04	8.0951E+08	1.1676E+08	1.1616E+04
CEC2017-F2	Best	1.9710E+101	5.0029E+150	2.0862E+124	1.2575E+91	2.8216E+84	2.3031E+121	2.0840E+115	**1.3414E+69**	5.7046E+79	5.9633E+111	6.5717E+91	2.6387E+71
	Worst	7.7078E+123	9.8321E+165	2.6152E+135	1.4990E+103	2.5639E+100	6.7651E+128	2.7049E+127	2.6462E+84	1.7656E+92	9.4550E+118	9.0444E+97	**4.6823E+79**
	Ave	1.5410E+123	1.9737E+165	5.2326E+134	2.9845E+102	5.1127E+99	1.3586E+128	5.4077E+126	5.3196E+83	3.5283E+91	1.8959E+118	1.8274E+97	**9.3614E+78**
	Std	3.4556E+123	6.7677E+299	1.1788E+135	6.6658E+102	1.1451E+100	3.0287E+128	1.2177E+127	1.1825E+84	7.8871E+91	4.2217E+118	4.0400E+97	**2.0932E+79**
CEC2017-F3	Best	2.7550E+05	8.5103E+05	2.7923E+05	1.6213E+05	2.1860E+05	5.2386E+05	3.6087E+05	**2.7785E+04**	3.0903E+04	4.0596E+05	1.8954E+05	1.1993E+05
	Worst	3.8569E+05	1.1293E+06	3.2082E+05	2.2309E+05	3.2610E+05	7.2176E+05	3.9391E+05	3.5678E+05	2.4883E+07	4.6073E+05	2.5090E+05	**1.7035E+05**
	Ave	3.3481E+05	1.0152E+06	2.9596E+05	1.9077E+05	2.5088E+05	6.1835E+05	3.8034E+05	**1.0681E+05**	5.0096E+06	4.3740E+05	2.1835E+05	1.4602E+05
	Std	4.3632E+04	1.1113E+05	1.5318E+04	2.2513E+04	4.4114E+04	7.9080E+04	**1.5320E+04**	1.4011E+05	1.1139E+07	2.0233E+04	2.3119E+04	2.1775E+04
CEC2017-F4	Best	2.6656E+03	7.6889E+03	2.0465E+03	8.0814E+02	8.1215E+02	2.2654E+03	6.1097E+03	5.8700E+02	6.3788E+02	1.5728E+03	7.5869E+02	**5.2428E+02**
	Worst	4.5337E+03	9.5180E+03	3.3320E+03	9.0886E+02	1.0392E+03	3.4896E+03	9.8011E+03	**7.6856E+02**	9.0322E+02	1.7148E+03	8.8560E+02	7.8745E+02
	Ave	3.5577E+03	8.4232E+03	2.7423E+03	8.5266E+02	9.4393E+02	2.9395E+03	7.4573E+03	7.0197E+02	7.3077E+02	1.6472E+03	8.0231E+02	**6.0151E+02**
	Std	7.4630E+02	7.4239E+02	5.2069E+02	**4.7578E+01**	9.8626E+01	4.9384E+02	1.4804E+03	6.8034E+01	1.0649E+02	5.7100E+01	5.1960E+01	9.9404E+01
CEC2017-F5	Best	1.0879E+03	1.5757E+03	1.5327E+03	1.2061E+03	1.2096E+03	1.3851E+03	1.3317E+03	**6.1427E+02**	6.8249E+02	1.6164E+03	1.2064E+03	7.5004E+02
	Worst	1.1611E+03	1.7781E+03	1.6184E+03	1.2951E+03	1.3403E+03	1.4669E+03	1.5990E+03	**7.2317E+02**	1.0945E+03	1.7094E+03	1.3435E+03	1.0756E+03
	Ave	1.1194E+03	1.6791E+03	1.5718E+03	1.2390E+03	1.2712E+03	1.4354E+03	1.4566E+03	**6.8587E+02**	8.1273E+02	1.6615E+03	1.2978E+03	9.2307E+02
	Std	**2.9688E+01**	7.3677E+01	2.9841E+01	3.6535E+01	6.2709E+01	3.2270E+01	1.1028E+02	4.2635E+01	1.6341E+02	3.5696E+01	5.8233E+01	1.1515E+02
CEC2017-F6	Best	6.3012E+02	6.8571E+02	6.8186E+02	6.5448E+02	6.5473E+02	6.6180E+02	6.6120E+02	**6.0140E+02**	6.0375E+02	6.6032E+02	6.0836E+02	**6.0167E+02**
	Worst	6.3506E+02	7.0245E+02	6.9109E+02	6.6052E+02	6.6487E+02	6.6956E+02	6.8907E+02	**6.0786E+02**	6.0867E+02	6.6889E+02	6.1487E+02	6.0808E+02
	Ave	6.3275E+02	6.9458E+02	6.8500E+02	6.5871E+02	6.6063E+02	6.6554E+02	6.7277E+02	6.0529E+02	6.0628E+02	6.6430E+02	6.1188E+02	**6.0333E+02**
	Std	**1.8809E+00**	6.9168E+00	3.8396E+00	2.2166E+00	4.0630E+00	3.0436E+00	1.0668E+01	2.1412E+00	2.0992E+00	3.2115E+00	2.3918E+00	3.1035E+00
CEC2017-F7	Best	1.7153E+03	3.4759E+03	3.5935E+03	1.1921E+03	2.0435E+03	3.1996E+03	2.5604E+03	1.0142E+03	**9.8673E+02**	2.2612E+03	1.6597E+03	1.1523E+03
	Worst	1.8716E+03	3.8205E+03	3.8770E+03	1.3011E+03	2.7746E+03	3.3690E+03	3.0924E+03	**1.1017E+03**	1.1721E+03	2.6743E+03	1.8380E+03	1.3536E+03
	Ave	1.7957E+03	3.5924E+03	3.7677E+03	1.2529E+03	2.5058E+03	3.2451E+03	2.8982E+03	**1.0615E+03**	1.0765E+03	2.5034E+03	1.7363E+03	1.2169E+03
	Std	7.4913E+01	1.3562E+02	1.0245E+02	4.3409E+01	2.8104E+02	6.8556E+01	2.0150E+02	**3.8424E+01**	7.9982E+01	1.5920E+02	7.2168E+01	8.0693E+01
CEC2017-F8	Best	1.3218E+03	2.0763E+03	1.9731E+03	1.6009E+03	1.4851E+03	1.6917E+03	1.6218E+03	**9.3992E+02**	9.8714E+02	1.9461E+03	1.5937E+03	1.1319E+03
	Worst	1.4632E+03	2.3481E+03	2.0639E+03	1.7525E+03	1.7160E+03	1.9638E+03	1.8318E+03	**9.8309E+02**	1.1247E+03	1.9884E+03	1.6415E+03	1.3025E+03
	Ave	1.3908E+03	2.1919E+03	2.0123E+03	1.6588E+03	1.5858E+03	1.8530E+03	1.7536E+03	**9.5865E+02**	1.0410E+03	1.9637E+03	1.6167E+03	1.1990E+03
	Std	6.4645E+01	9.9911E+01	3.6630E+01	6.4116E+01	1.0444E+02	1.1720E+02	9.4107E+01	1.9051E+01	5.0360E+01	**1.4171E+01**	2.4932E+01	7.1024E+01
CEC2017-F9	Best	1.7887E+04	5.2810E+04	5.0758E+04	7.9299E+03	1.9610E+04	3.4490E+04	3.8249E+04	2.6093E+03	**2.5916E+03**	4.4211E+04	3.3696E+03	3.4458E+03
	Worst	4.2561E+04	1.0746E+05	5.5497E+04	1.1211E+04	2.8177E+04	4.8110E+04	6.4823E+04	5.2763E+03	**4.3655E+03**	5.9068E+04	1.1706E+04	4.9529E+03
	Ave	3.1159E+04	7.5487E+04	5.3532E+04	9.7570E+03	2.3921E+04	3.9120E+04	5.1119E+04	4.0205E+03	**3.3387E+03**	5.3051E+04	6.1617E+03	4.2847E+03
	Std	9.1158E+03	2.2274E+04	1.8568E+03	1.3604E+03	3.8038E+03	5.5965E+03	1.0240E+04	1.0852E+03	6.8152E+02	6.0079E+03	3.2783E+03	**6.1553E+02**
CEC2017-F10	Best	1.4998E+04	2.5379E+04	2.0690E+04	1.4789E+04	1.5186E+04	1.9421E+04	3.0175E+04	1.2108E+04	2.5918E+04	2.5728E+04	2.5217E+04	**1.2047E+04**
	Worst	1.7720E+04	2.7353E+04	2.5941E+04	1.8405E+04	1.8699E+04	2.5103E+04	3.2637E+04	**1.4803E+04**	3.0299E+04	2.7828E+04	2.8532E+04	1.6198E+04
	Ave	1.6440E+04	2.6572E+04	2.2828E+04	1.5989E+04	1.6416E+04	2.2706E+04	3.1582E+04	1.3379E+04	2.8360E+04	2.6885E+04	2.7320E+04	**1.2811E+04**
	Std	1.2749E+03	**8.2324E+02**	2.1611E+03	1.4588E+03	1.4140E+03	2.4843E+03	9.3454E+02	9.5557E+02	1.5701E+03	8.3848E+02	1.3654E+03	1.7386E+03
CEC2017-F11	Best	5.4423E+04	1.2366E+05	3.6653E+04	4.5371E+03	1.5184E+04	1.4478E+05	7.6916E+04	2.3076E+03	2.5007E+03	6.7305E+04	6.1974E+03	**2.0405E+03**
	Worst	7.1648E+04	1.6611E+05	6.8537E+04	9.0851E+03	4.7170E+04	2.2225E+05	9.3591E+04	**2.6669E+03**	3.3616E+03	9.8967E+04	1.0130E+04	3.1283E+03
	Ave	6.3984E+04	1.3927E+05	5.2886E+04	6.7089E+03	2.4826E+04	1.8843E+05	8.4385E+04	2.4953E+03	2.9375E+03	8.7820E+04	8.0525E+03	**2.1322E+03**
	Std	6.1939E+03	2.0248E+04	1.4746E+04	1.6618E+03	1.2816E+04	3.7089E+04	6.0941E+03	**1.6364E+02**	3.7038E+02	1.3517E+04	1.5049E+03	4.4123E+02
CEC2017-F12	Best	2.1877E+09	6.8893E+09	9.8726E+08	6.8933E+06	5.2137E+07	1.0761E+09	4.1424E+09	1.1814E+07	**1.7233E+06**	2.8993E+08	5.5704E+07	3.0252E+06
	Worst	6.3195E+09	8.9405E+09	1.8666E+09	2.1528E+07	1.4363E+08	4.4815E+09	9.3053E+09	4.1653E+07	1.6675E+07	4.8893E+08	1.2451E+08	**7.8468E+06**
	Ave	3.3574E+09	7.8873E+09	1.3633E+09	1.4074E+07	8.5633E+07	2.3232E+09	6.2378E+09	2.2437E+07	7.6827E+06	3.7013E+08	7.8658E+07	**5.7317E+06**
	Std	1.7083E+09	7.3942E+08	3.1917E+08	6.4379E+06	4.1194E+07	1.3651E+09	2.0874E+09	1.1954E+07	7.7619E+06	7.8956E+07	2.7911E+07	**1.9264E+06**
CEC2017-F13	Best	1.2179E+08	1.0194E+08	1.1443E+07	1.1912E+04	1.0544E+04	3.9360E+05	4.4766E+07	1.2702E+04	6.1620E+03	5.0653E+04	3.3506E+04	**6.0390E+03**
	Worst	4.3954E+08	2.4352E+08	1.8772E+07	1.6768E+04	4.3292E+05	1.7502E+07	1.6714E+08	2.2346E+04	**1.0907E+04**	1.3411E+05	8.0902E+04	1.1352E+04
	Ave	2.5423E+08	1.6200E+08	1.6288E+07	1.4783E+04	1.0880E+05	4.1409E+06	1.0435E+08	1.8179E+04	9.3287E+03	1.0417E+05	5.1897E+04	**8.7705E+03**
	Std	1.3413E+08	5.4987E+07	2.7883E+06	**1.7410E+03**	1.8237E+05	7.4683E+06	4.5094E+07	4.3790E+03	1.9179E+03	3.2441E+04	1.8526E+04	2.1785E+03
CEC2017-F14	Best	2.4533E+06	7.1118E+06	4.8438E+06	5.4238E+05	7.9736E+05	3.4014E+06	1.2009E+06	1.2791E+04	5.0847E+03	4.1971E+06	3.3216E+04	**4.4982E+03**

(*continued on next page*)

Table 6 (*continued*)

ID	index	GWO	WOA	HHO	IPO	MSMA	COA	OMA	LSHADE _cnEpSin	LSHADE _SPACMA	MadDE	EA4eig	BHO
	Worst	5.4185E+06	2.2166E+07	6.8145E+06	8.6327E+05	3.5340E+06	4.9547E+06	1.9393E+06	**2.6595E+04**	5.1549E+04	7.4985E+06	3.2088E+05	6.3461E+04
	Ave	4.0857E+06	1.1690E+07	5.6369E+06	6.4060E+05	1.4156E+06	4.2149E+06	1.5827E+06	1.9820E+04	**1.6009E+04**	5.5652E+06	1.2034E+05	3.2120E+04
	Std	1.2616E+06	6.0744E+06	7.7256E+05	1.3679E+05	1.1963E+06	5.9799E+05	3.1955E+05	**5.8538E+03**	1.9953E+04	1.2275E+06	1.2167E+05	2.5279E+04
CEC2017-F15	Best	1.4333E+07	4.4250E+06	1.8359E+06	4.6848E+03	3.7247E+03	1.6832E+05	4.2960E+05	4.3603E+03	2.6952E+03	1.6554E+04	1.9243E+04	**2.6077E+03**
	Worst	6.3358E+07	3.8560E+07	3.5181E+06	9.4317E+03	1.8880E+05	8.5882E+06	2.5189E+06	1.1011E+04	**5.7016E+03**	4.9870E+04	4.1920E+04	6.9852E+03
	Ave	4.1216E+07	2.4557E+07	2.8012E+06	6.2198E+03	4.3006E+04	2.0415E+06	1.0408E+06	7.2707E+03	**4.5067E+03**	3.1058E+04	3.0141E+04	4.5322E+03
	Std	1.7591E+07	1.3746E+07	6.6871E+05	1.9022E+03	8.1379E+04	3.6794E+06	8.4590E+05	2.8606E+03	**1.4532E+03**	1.3646E+04	8.3739E+03	1.7692E+03
CEC2017-F16	Best	5.4024E+03	1.1826E+04	7.9297E+03	5.7753E+03	4.8613E+03	5.4420E+03	6.5056E+03	**3.7116E+03**	7.5033E+03	8.2436E+03	8.3059E+03	4.2352E+03
	Worst	5.9985E+03	1.6131E+04	9.1718E+03	7.0349E+03	7.1963E+03	8.0596E+03	9.7936E+03	**4.5442E+03**	9.7997E+03	9.4027E+03	8.9309E+03	5.6541E+03
	Ave	5.6825E+03	1.4232E+04	8.5499E+03	6.3469E+03	6.0182E+03	7.1136E+03	8.3975E+03	**4.2088E+03**	8.4326E+03	8.8879E+03	8.5903E+03	4.8637E+03
	Std	2.6254E+02	1.6226E+03	5.3355E+02	5.1544E+02	9.4116E+02	1.0673E+03	1.2362E+03	3.1090E+02	1.0778E+03	4.8344E+02	**2.2976E+02**	6.2089E+02
CEC2017-F17	Best	3.7907E+03	7.8454E+03	5.6315E+03	4.5477E+03	5.7128E+03	5.4962E+03	5.3784E+03	**3.5860E+03**	4.7195E+03	5.4258E+03	6.1842E+03	3.7058E+03
	Worst	5.2453E+03	1.0728E+04	8.3851E+03	5.5440E+03	7.8874E+03	6.5897E+03	6.4668E+03	**4.2711E+03**	7.1102E+03	6.3501E+03	6.8512E+03	5.0970E+03
	Ave	4.3303E+03	9.1782E+03	6.7345E+03	4.9607E+03	6.3501E+03	6.0614E+03	5.8224E+03	**3.9432E+03**	6.0984E+03	5.8599E+03	6.4313E+03	4.3212E+03
	Std	5.4422E+02	1.2467E+03	1.1316E+03	4.0960E+02	9.3328E+02	4.7969E+02	4.4554E+02	**2.6332E+02**	8.7167E+02	3.3748E+02	3.1418E+02	5.6021E+02
CEC2017-F18	Best	4.6483E+06	7.1999E+06	2.8903E+06	5.6857E+05	1.1563E+06	4.5656E+06	1.8507E+06	5.8934E+04	**2.8325E+04**	3.5396E+06	1.2192E+05	9.2222E+04
	Worst	1.8603E+07	1.7731E+07	6.1953E+06	9.3212E+05	1.0924E+07	1.3369E+07	6.1304E+06	1.5667E+05	**6.6620E+04**	5.6534E+06	5.9070E+05	2.6193E+05
	Ave	8.6871E+06	1.2571E+07	4.5163E+06	7.7381E+05	5.1099E+06	7.0414E+06	2.9209E+06	9.9628E+04	**4.5670E+04**	4.6246E+06	3.1766E+05	1.7224E+05
	Std	5.7893E+06	3.7313E+06	1.3819E+06	1.3689E+05	3.6156E+06	3.6836E+06	1.8159E+06	3.6778E+04	**1.4232E+04**	8.5949E+05	2.2000E+05	7.3537E+04
CEC2017-F19	Best	1.4132E+07	1.4685E+07	5.2666E+06	3.1503E+03	2.4470E+03	4.7684E+05	1.3438E+06	2.4410E+03	2.2326E+03	1.0333E+04	1.5718E+04	**2.1731E+03**
	Worst	2.9570E+08	1.5259E+08	2.4367E+07	4.9115E+03	3.2050E+04	5.6383E+06	3.3729E+06	7.7467E+03	**3.9890E+03**	2.7985E+04	7.4425E+04	4.1163E+03
	Ave	9.7791E+07	7.8750E+07	1.5385E+07	4.1410E+03	1.1544E+04	2.6453E+06	2.2783E+06	5.4710E+03	3.0377E+03	1.9471E+04	3.7255E+04	**2.8330E+03**
	Std	1.1826E+08	5.4486E+07	6.8913E+06	**7.0799E+02**	1.1700E+04	2.0786E+06	8.6765E+05	2.1658E+03	7.9099E+02	6.8240E+03	2.2834E+04	8.2184E+02
CEC2017-F20	Best	4.2815E+03	5.5342E+03	5.4181E+03	4.4336E+03	4.0895E+03	6.0881E+03	6.9682E+03	**3.6180E+03**	4.0315E+03	5.5067E+03	6.4349E+03	3.6809E+03
	Worst	8.0879E+03	7.4224E+03	6.2839E+03	5.7302E+03	6.1933E+03	7.1227E+03	7.5408E+03	**4.2579E+03**	7.9886E+03	6.1796E+03	7.0103E+03	4.4926E+03
	Ave	5.8177E+03	6.6520E+03	5.7482E+03	5.0104E+03	5.1611E+03	6.4820E+03	7.2222E+03	3.9240E+03	6.8199E+03	5.8868E+03	6.6275E+03	**3.7262E+03**
	Std	1.8242E+03	7.7112E+02	3.5690E+02	5.6519E+02	8.3033E+02	4.8649E+02	2.5963E+02	2.8227E+02	1.6190E+03	2.7444E+02	**2.3539E+02**	4.3896E+02
CEC2017-F21	Best	2.8852E+03	4.0816E+03	4.0944E+03	2.9249E+03	2.9874E+03	3.0225E+03	3.0988E+03	**2.4804E+03**	2.4843E+03	3.3088E+03	3.0903E+03	2.5858E+03
	Worst	3.0886E+03	4.3988E+03	4.3541E+03	3.0775E+03	3.2141E+03	3.5830E+03	3.3693E+03	**2.5359E+03**	2.6611E+03	3.4210E+03	3.1586E+03	2.7750E+03
	Ave	2.9887E+03	4.2587E+03	4.2346E+03	2.9757E+03	3.0886E+03	3.3458E+03	3.2081E+03	**2.5221E+03**	2.5985E+03	3.3601E+03	3.1364E+03	2.7148E+03
	Std	8.7443E+01	1.2187E+02	1.1292E+02	5.8007E+01	8.9486E+01	2.1420E+02	1.1066E+02	**2.2961E+01**	6.7699E+01	4.4516E+01	2.5619E+01	7.3075E+01
CEC2017-F22	Best	1.8669E+04	2.6329E+04	2.4532E+04	2.0290E+04	1.5599E+04	2.3926E+04	3.2805E+04	1.4792E+04	3.2043E+04	2.7304E+04	2.8384E+04	**1.4607E+04**
	Worst	3.3578E+04	3.1906E+04	2.6621E+04	2.1787E+04	1.7926E+04	3.0836E+04	3.4482E+04	**1.6168E+04**	3.8824E+04	3.0961E+04	3.1583E+04	1.7246E+04
	Ave	2.2175E+04	2.9522E+04	2.5304E+04	2.0892E+04	1.7374E+04	2.8007E+04	3.3928E+04	1.5384E+04	3.6524E+04	2.9122E+04	2.9737E+04	**1.4478E+04**
	Std	6.3974E+03	2.4631E+03	8.0628E+02	6.5696E+02	1.0481E+03	2.8630E+03	5.9241E+02	**5.1279E+02**	2.7042E+03	1.3613E+03	1.3309E+03	1.1707E+03
CEC2017-F23	Best	3.3980E+03	4.5559E+03	5.0759E+03	3.6812E+03	3.2810E+03	3.8447E+03	3.7826E+03	**3.0092E+03**	3.4114E+03	3.7382E+03	3.6202E+03	3.2650E+03
	Worst	3.5890E+03	5.3335E+03	5.6695E+03	4.0778E+03	3.4015E+03	4.3499E+03	4.0352E+03	**3.0357E+03**	3.7953E+03	3.8975E+03	3.6704E+03	3.4721E+03
	Ave	3.5254E+03	4.9967E+03	5.4119E+03	3.8814E+03	3.3653E+03	4.0898E+03	3.8767E+03	**3.0117E+03**	3.5664E+03	3.8257E+03	3.6579E+03	3.3655E+03
	Std	7.5517E+01	3.1248E+02	2.7858E+02	1.7029E+02	4.9130E+01	1.8288E+02	9.8424E+01	**1.2345E+01**	1.5547E+02	6.1279E+01	2.0358E+01	8.5809E+01
CEC2017-F24	Best	3.8167E+03	5.7997E+03	6.2346E+03	4.2038E+03	3.8786E+03	4.4454E+03	4.6437E+03	**3.5320E+03**	3.8487E+03	4.3317E+03	4.0957E+03	3.6323E+03
	Worst	4.2858E+03	6.9127E+03	7.6109E+03	4.5428E+03	4.2642E+03	4.9216E+03	5.2419E+03	**3.7529E+03**	4.0575E+03	4.4610E+03	4.2104E+03	3.8630E+03
	Ave	4.0459E+03	6.5815E+03	7.0861E+03	4.3917E+03	4.1288E+03	4.7182E+03	4.9324E+03	**3.6042E+03**	3.9456E+03	4.3811E+03	4.1623E+03	3.7252E+03
	Std	1.8985E+02	4.5667E+02	5.7873E+02	1.3706E+02	1.4984E+02	2.1480E+02	2.8776E+02	9.0277E+01	7.8702E+01	4.8519E+01	**4.5092E+01**	1.0149E+02
CEC2017-F25	Best	5.2774E+03	6.0537E+03	4.7002E+03	3.5253E+03	3.5798E+03	4.8732E+03	6.8816E+03	**3.3008E+03**	3.4263E+03	4.3129E+03	3.4852E+03	3.3391E+03
	Worst	5.9583E+03	7.4080E+03	5.6022E+03	3.6627E+03	3.7602E+03	6.2116E+03	8.4319E+03	**3.4673E+03**	3.5472E+03	4.7268E+03	3.6459E+03	3.4853E+03
	Ave	5.6092E+03	6.7118E+03	4.9977E+03	3.5814E+03	3.6384E+03	5.4639E+03	7.5385E+03	**3.3862E+03**	3.4902E+03	4.5232E+03	3.5619E+03	3.4375E+03
	Std	2.6207E+02	6.6544E+02	3.6455E+02	6.4301E+01	7.6649E+01	4.8647E+02	7.4069E+02	5.4192E+01	**4.6359E+01**	1.6215E+02	6.7211E+01	5.6524E+01
CEC2017-F26	Best	1.3089E+04	2.9497E+04	2.5969E+04	1.9948E+04	1.5597E+04	2.5406E+04	2.1817E+04	8.1917E+03	8.4467E+03	2.2623E+04	1.3861E+04	**8.0587E+03**
	Worst	1.5738E+04	3.7692E+04	3.1966E+04	2.3017E+04	1.8359E+04	3.0958E+04	2.7987E+04	9.0446E+03	1.3655E+04	2.4105E+04	1.5206E+04	**8.7811E+03**
	Ave	1.4354E+04	3.3003E+04	2.8795E+04	2.0860E+04	1.6441E+04	2.8268E+04	2.4314E+04	8.5793E+03	1.0783E+04	2.3398E+04	1.4509E+04	**8.4038E+03**
	Std	1.1289E+03	3.1855E+03	2.1739E+03	1.3149E+03	1.1332E+03	2.3491E+03	2.4503E+03	3.7007E+02	2.4542E+03	6.5606E+02	6.2753E+02	**3.3066E+02**
CEC2017-F27	Best	3.7907E+03	4.6494E+03	4.6446E+03	4.6117E+03	3.4750E+03	3.9917E+03	4.3714E+03	3.5035E+03	3.4875E+03	4.0231E+03	**3.2079E+03**	3.4510E+03
	Worst	4.0923E+03	6.2349E+03	6.2076E+03	5.1065E+03	3.6301E+03	4.2103E+03	4.9957E+03	3.6410E+03	3.5637E+03	4.1154E+03	**3.5436E+03**	3.5984E+03
	Ave	3.9428E+03	5.5926E+03	5.4820E+03	4.8545E+03	3.5798E+03	4.1427E+03	4.7045E+03	3.5918E+03	3.5210E+03	4.0758E+03	**3.3667E+03**	3.5244E+03

(*continued on next page*)

Table 6 (*continued*)

ID	index	GWO	WOA	HHO	IPO	MSMA	COA	OMA	LSHADE_cnEpSin	LSHADE_SPACMA	MadDE	EA4eig	BHO
	Std	1.3596E+02	6.2296E+02	6.8228E+02	2.2640E+02	7.1179E+01	8.8517E+01	2.5198E+02	5.5133E+01	3.5656E+01	**3.3214E+01**	1.5353E+02	4.9452E+01
CEC2017-F28	Best	5.5979E+03	8.0425E+03	5.3454E+03	3.6925E+03	3.6299E+03	5.4775E+03	9.9759E+03	3.4768E+03	3.8559E+03	4.9303E+03	3.5882E+03	**3.4728E+03**
	Worst	9.3965E+03	1.0408E+04	6.0247E+03	3.7869E+03	3.7219E+03	6.8920E+03	1.2608E+04	**3.5273E+03**	5.3365E+04	6.0070E+03	3.8879E+03	3.5396E+03
	Ave	7.2936E+03	8.8343E+03	5.7066E+03	3.7490E+03	3.6650E+03	6.2860E+03	1.1332E+04	**3.4916E+03**	4.1120E+04	5.5262E+03	3.6827E+03	3.5035E+03
	Std	1.6334E+03	1.0294E+03	3.0014E+02	3.0566E+01	4.0838E+01	5.7301E+02	1.1483E+03	**1.9402E+01**	2.1071E+04	4.2751E+02	1.2364E+02	2.8568E+01
CEC2017-F29	Best	7.2234E+03	1.5020E+04	9.9153E+03	7.4043E+03	7.2180E+03	8.4046E+03	9.4110E+03	**5.0965E+03**	7.8122E+03	9.2140E+03	8.2549E+03	5.8505E+03
	Worst	8.4459E+03	1.7851E+04	1.1143E+04	8.9146E+03	8.8668E+03	1.0784E+04	1.0396E+04	**5.9439E+03**	1.0955E+04	1.0222E+04	8.8597E+03	7.4446E+03
	Ave	7.8623E+03	1.6805E+04	1.0673E+04	8.2290E+03	7.9020E+03	9.4303E+03	9.9789E+03	**5.4517E+03**	9.2078E+03	9.7352E+03	8.5830E+03	6.5952E+03
	Std	5.6554E+02	1.0523E+03	5.0274E+02	5.3875E+02	6.6721E+02	1.0267E+03	3.2769E+02	4.3914E+02	1.3179E+03	3.8237E+02	**2.2355E+02**	6.5710E+02
CEC2017-F30	Best	1.4225E+08	6.8578E+08	7.5479E+07	6.2191E+04	5.4344E+04	3.3024E+07	2.8428E+07	7.7583E+04	3.3619E+04	3.1540E+06	9.9938E+05	**2.6903E+04**
	Worst	7.3380E+08	1.4239E+09	1.8859E+08	6.1020E+05	3.5138E+05	6.1385E+07	2.1293E+08	2.5405E+05	1.3302E+05	8.6034E+06	1.5066E+06	**1.2273E+05**
	Ave	3.1940E+08	9.6092E+08	1.1228E+08	2.8257E+05	1.6742E+05	4.5753E+07	9.0107E+07	1.4021E+05	7.9306E+04	5.6283E+06	1.1820E+06	**6.3369E+04**
	Std	2.3855E+08	3.0276E+08	4.5104E+07	2.2622E+05	1.1202E+05	1.2647E+07	7.0707E+07	6.8964E+04	**4.4195E+04**	2.0485E+06	1.9022E+05	4.4965E+04
average Best	6.933	11.233	9.600	5.200	4.933	8.700	9.433	2.100	3.500	8.367	6.100	**1.900**	
rank													
average Worst rank	7.500	11.167	9.133	4.833	5.767	8.767	9.400	**1.933**	4.667	7.533	5.167	2.133	
average Ave rank	7.033	11.300	9.367	4.933	5.467	8.700	9.233	2.000	4.400	7.967	5.733	**1.867**	
average Std rank	8.133	10.467	7.833	4.667	7.200	8.733	8.300	**2.933**	5.633	5.300	4.200	4.600	

the top three, gradually widening its gap with LSHADE_SPACMA (0.4–1.9334) while closing in on LSHADE_cnEpSin (1.2333–0.4667); In the third stage, the final convergence phase, BHO surpassed LSHADE_cnEpSin at approximately 480D, ranking first and demonstrating strong capability in escaping local convergence. Overall, BHO quickly established an early advantage with near-optimal descent rates and maintained robust convergence throughout, highlighting its superior global optimization capability and robustness.

To examine the differences between the BHO algorithm and other algorithms, this study employed the Wilcoxon rank-sum test [51] on the mean results of 30 independent runs. The outcomes are reported in Table 7. According to this test, a P-value less than 0.05 indicates a statistically significant difference between the two algorithms. For clarity of comparison, a "+" is used to indicate that BHO is significantly superior to the comparison algorithm, while a "-" denotes otherwise. Results with P-values greater than 0.05 are highlighted in bold within the Table 7. The results show that BHO exhibits no significant difference compared with LSHADE_cnEpSin on F13, F14, and F30, and with MSMA on F22 and F23, while all other functions passed the significance test. These findings indicate that the BHO algorithm is significantly superior to other comparison algorithms in most test functions, further confirming its strong global optimization capability.

4.4. Running time evaluation

In order to evaluate the computational efficiency of the algorithm, this paper calculates the average CPU running time of each method on the CEC2017 ($D = 100$). Table 4 summarizes these results, with the final row providing the mean values across all functions. All algorithms followed the identical experimental environment and stopping conditions specified in Section 3. To ensure comparability, the measured time includes both the main optimization loop and the objective function evaluations. In general, the running time of classical swarm intelligence methods such as GWO and WOA is kept at a low level because each generation mainly involves simple location update and boundary processing. The two LSHADE variants also show high time efficiency, which indicates that the additional computational overhead can be controlled within an acceptable range by reasonable operator design and adaptive control in the differential evolution framework.

As indicated by the statistics in Table 4, BHO exhibits a medium-to-high average runtime. It is slower than lightweight algorithms and some methods of moderate complexity, yet it is markedly faster than the more time-consuming EA4eig and is comparable to MSMA. This trend is consistent with the algorithmic design of BHO. Beyond the baseline position update, BHO incorporates archive maintenance, the distance-/dominance-based Explorpolis acceptance test, back-off candidate handling, and group migration. These components introduce additional filtering, sorting, and distance computations, thereby increasing the per-iteration constant overhead.

To quantify the cross-function variability of runtime, we further compute the coefficient of variation (CV), defined as the ratio of the standard deviation s_t to the mean $\bar{t}$, i.e., $\mathrm{CV} = s_t/\bar{t}$. Here, $\bar{t}$ and s_t are calculated from the per-function runtimes on F1–F30 reported in Table 11. BHO shows a more compact runtime distribution across the test suite, achieving a 73% reduction in CV compared with MSMA and a 28% reduction compared with EA4eig. This consistency suggests that BHO is less sensitive to function complexity. Its stable runtime improves predictability and simplifies resource planning for diverse problems. Overall, the runtime overhead of BHO primarily stems from the auxiliary mechanisms introduced to strengthen search control and robustness, while remaining stable and predictable in practice.

Table 7
Wilcoxon rank-sum test results for each algorithm ($D = 100$).

ID	GWO	WOA	HHO	IPO	MSMA	COA	OMA	LSHADE_cnEpSin	LSHADE_SPACMA	MadDE	EA4eig
F1	1.4927E-126	2.1505E-128	3.4492E-133	8.9785E-23	1.4931E-60	4.8922E-130	8.1197E-140	4.0767E-112	1.4065E-114	3.7579E-120	2.9032E-69
F2	2.4088E-103	3.0934E-160	3.2395E-146	1.6709E-68	8.6671E-62	1.0167E-132	2.6497E-113	6.8349E-04	3.0802E-04	3.2733E-112	1.1640E-72
F3	4.6556E-151	2.0750E-161	5.3559E-136	1.8747E-73	1.3590E-96	2.3280E-161	2.5427E-155	3.1899E-55	2.2631E-162	1.0213E-157	3.3082E-81
F4	1.8894E-124	2.4503E-135	2.5773E-139	6.0118E-48	8.5708E-65	3.6824E-129	2.3367E-138	7.5749E-130	8.8404E-93	8.0655E-120	1.9667E-35
F5	1.6926E-115	1.3255E-136	1.1600E-138	6.6397E-89	9.5885E-103	1.9033E-121	1.4873E-131	1.1636E-117	3.9438E-125	4.6011E-140	4.8566E-114
F6	2.2826E-116	1.8329E-144	4.4884E-143	1.0774E-129	1.3375E-130	1.7207E-129	1.1857E-133	2.5058E-113	4.0858E-138	1.2938E-131	2.2524E-39
F7	4.0509E-115	1.3084E-153	1.4435E-157	2.2175E-25	4.3451E-127	2.8546E-147	4.1159E-147	1.0826E-90	6.0140E-106	1.3215E-130	1.2292E-107
F8	1.5926E-115	1.9095E-143	1.4484E-144	1.8953E-111	3.9750E-110	5.8150E-129	1.1619E-129	4.2066E-106	1.9701E-69	1.3918E-138	1.0022E-116
F9	6.6237E-124	4.4828E-146	3.7460E-134	2.4771E-93	5.8203E-131	6.3439E-127	2.5172E-130	2.9833E-48	5.2837E-105	2.8475E-136	1.1973E-80
F10	9.0900E-106	5.0642E-108	4.3631E-109	5.4005E-17	5.9068E-53	4.7093E-86	1.9464E-141	8.1911E-45	1.7512E-127	8.9534E-119	2.7561E-122
F11	2.1490E-110	2.1191E-142	1.6223E-142	2.2055E-62	2.2866E-90	6.4465E-158	1.3815E-131	7.5823E-131	1.2062E-114	7.9589E-131	2.7303E-61
F12	2.3635E-124	7.8259E-129	3.4222E-136	1.2969E-35	9.7267E-71	3.0989E-128	3.0594E-133	**2.9759E-01**	1.2831E-30	2.3845E-109	9.5629E-84
F13	5.6625E-132	7.7066E-122	1.3667E-133	7.0499E-26	9.9565E-75	1.3108E-109	1.6351E-133	**5.1207E-01**	9.9738E-104	1.1095E-99	3.7090E-90
F14	1.5875E-129	2.9545E-142	1.0428E-146	5.8469E-107	1.3539E-109	1.3851E-134	7.0711E-115	2.0098E-149	4.3783E-68	3.7376E-133	1.4950E-91
F15	9.8691E-137	4.9993E-126	2.8475E-136	4.5113E-22	2.2825E-91	3.9627E-120	2.5802E-126	9.7519E-17	2.8462E-110	5.1842E-103	5.4545E-102
F16	4.1742E-113	3.3124E-153	1.0795E-149	1.5005E-104	9.6872E-88	2.2855E-120	8.2423E-129	1.4048E-93	3.8996E-132	1.2061E-135	2.6686E-132
F17	4.5850E-76	4.4324E-143	1.1328E-147	1.1471E-82	2.4585E-126	3.5495E-125	2.1919E-122	6.2383E-114	2.2221E-116	4.1518E-128	2.3229E-132
F18	1.4403E-143	2.6765E-140	3.1283E-137	2.6340E-99	2.7793E-132	9.8801E-140	1.0553E-122	1.2568E-153	1.1952E-149	3.3516E-131	9.2175E-94
F19	9.8301E-139	1.6938E-131	1.0238E-140	3.8337E-09	9.3599E-16	2.3111E-112	6.6323E-125	1.9300E-136	1.3401E-143	2.8146E-76	2.9083E-83
F20	1.2030E-128	1.3034E-117	5.3940E-105	1.3957E-66	4.8751E-09	1.3766E-127	1.9439E-138	1.7231E-117	4.7406E-121	3.0773E-113	2.9536E-131
F21	9.2376E-120	1.0350E-154	1.0274E-157	3.8968E-79	2.6332E-103	1.8431E-130	1.9446E-122	2.2607E-94	2.1378E-76	1.1008E-129	4.0403E-111
F22	6.5110E-133	2.4221E-123	4.0123E-121	4.7201E-14	**2.7622E-01**	2.3837E-123	2.5505E-148	1.1138E-81	2.0228E-179	4.3536E-129	2.0518E-134
F23	9.3844E-114	1.2908E-150	3.0037E-159	9.5647E-122	**5.8508E-01**	4.5437E-130	8.8876E-129	7.3116E-126	3.2160E-76	8.5082E-124	1.2376E-110
F24	2.1128E-112	4.6308E-149	1.7107E-156	6.1280E-112	1.1845E-90	8.8426E-124	1.8158E-133	5.3687E-120	2.1535E-38	1.7398E-120	1.5998E-101
F25	7.6650E-115	1.9951E-121	3.4282E-132	5.4814E-46	1.6139E-58	3.4882E-122	2.6693E-136	9.9823E-133	2.3364E-13	9.2676E-117	1.9250E-55
F26	2.6784E-121	2.0445E-147	3.6162E-151	3.0575E-131	1.8299E-113	8.9941E-143	2.5972E-138	3.2932E-104	1.4181E-17	3.6153E-137	3.3405E-104
F27	1.4501E-122	1.4209E-145	1.7460E-156	2.5771E-138	1.6167E-53	2.7310E-125	5.2800E-139	6.2530E-51	5.5598E-160	7.6585E-131	3.6298E-02
F28	4.0698E-116	1.1164E-122	4.8896E-129	3.9066E-55	6.2135E-35	2.0206E-122	8.7033E-137	3.2444E-101	4.7436E-184	5.7779E-117	2.7499E-61
F29	4.3660E-112	3.1582E-145	1.3030E-143	8.5207E-104	5.6127E-84	4.4846E-123	1.7801E-128	3.7106E-122	8.6530E-112	8.7178E-127	7.0933E-115
F30	6.7134E-134	1.6383E-135	1.5086E-141	7.7991E-62	7.5170E-23	2.5887E-114	1.6147E-127	**3.6401E-01**	4.5058E-35	2.2748E-106	7.7272E-80
±	0/30	0/30	0/30	0/30	2/28	0/30	0/30	3/27	0/30	0/30	0/30

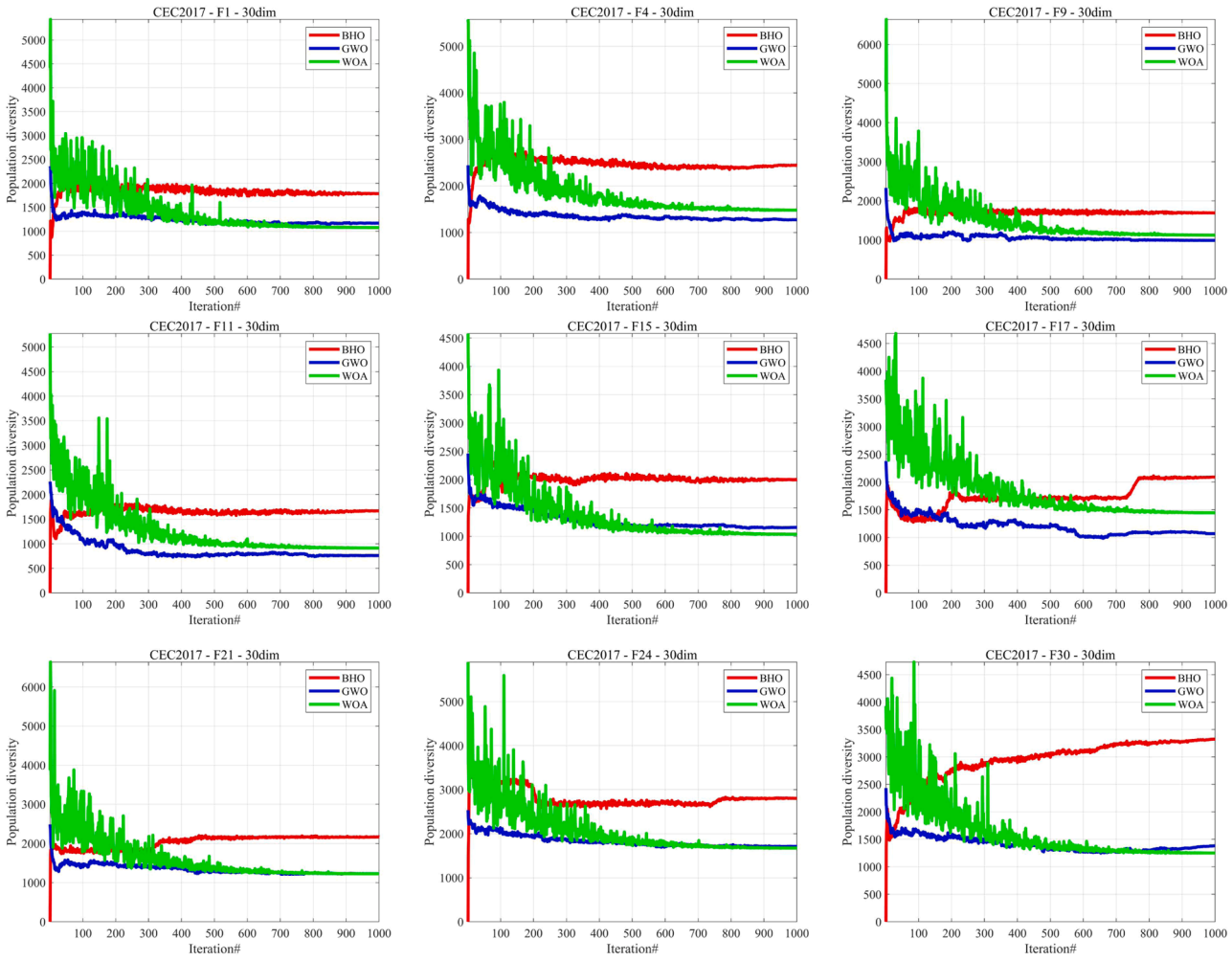


Fig. 9. Population diversity of BHO, GWO, and WOA.

4.5. Population diversity analysis

To examine population diversity during the search process, experiments are conducted on the CEC2017 benchmark suite with a 30-dimensional setting. We select GWO and WOA as baselines due to their representative and classical search paradigms: GWO adopts a contraction-oriented update guided by the $\alpha/\beta/\delta$ leaders (leader-driven attraction and aggregation), whereas WOA achieves the exploration–exploitation transition via probabilistic switching among random search, encircling, and spiral attacking behaviors (stochastic perturbation with local intensification). Both methods typically maintain relatively high dispersion in early iterations, making them suitable high-diversity references.

The diversity is quantified by the inertia moment I_C defined in Eq. (19), which measures the spread of search agents around the center of mass. Here, $x_i^d(t)$ denotes the position of the i-th agent in the d-th dimension at iteration t. Moreover, $c_d(t)$ describes the population dispersion relative to its center of mass $c(t)$ along each dimension, as given in Eq. (20) [11].

$$I_C(t) = \sqrt{\sum_{i=1}^{n}\sum_{d=1}^{D}\left(x_{id}(t) - c_d(t)\right)^2}, \tag{19}$$

$$c_d(t) = \frac{1}{D}\sum_{i=1}^{n} x_{id}(t). \tag{20}$$

Fig. 9 presents the comparative results of population diversity for BHO, WOA, and GWO, where diversity is measured by the inertia-moment index I_C. On CEC2017 with 30 dimensions, the I_C-based curves of GWO and WOA generally decrease with iterations, which is consistent with their typical convergence behavior—i.e., the population progressively aggregates under leader-driven guidance and gradually shifts into the exploitation phase. In particular, WOA exhibits more pronounced oscillations, indicating that the probabilistic strategy switching and spiral updates introduce stronger stochastic perturbations, leading to larger fluctuations in the population distribution during the early-to-middle stages. Eventually, the diversity level of WOA approaches that of GWO and becomes stable.

By contrast, BHO shows a non-monotonic trajectory of I_C, characterized by a "low-at-start, then increasing, and finally stabilizing" pattern. The relatively low initial diversity mainly stems from the structured initialization (generated around cluster centroids), which yields a more compact initial distribution. As the search progresses, Explorer updates, the Explorpolis criterion, and group migration prevent premature crowding. These mechanisms expand the search area and drive diversity recovery, maintaining higher dispersion in mid-to-late iterations. Overall, BHO preserves higher and more stable population dispersion while ensuring convergence progress, achieving the highest diversity among the three methods and indicating stronger sustained global exploration and improved resistance to premature convergence.

5. Multi-UAV mec-pp task

Multi-UAV MEC-PP optimizes the flight paths of UAVs equipped with

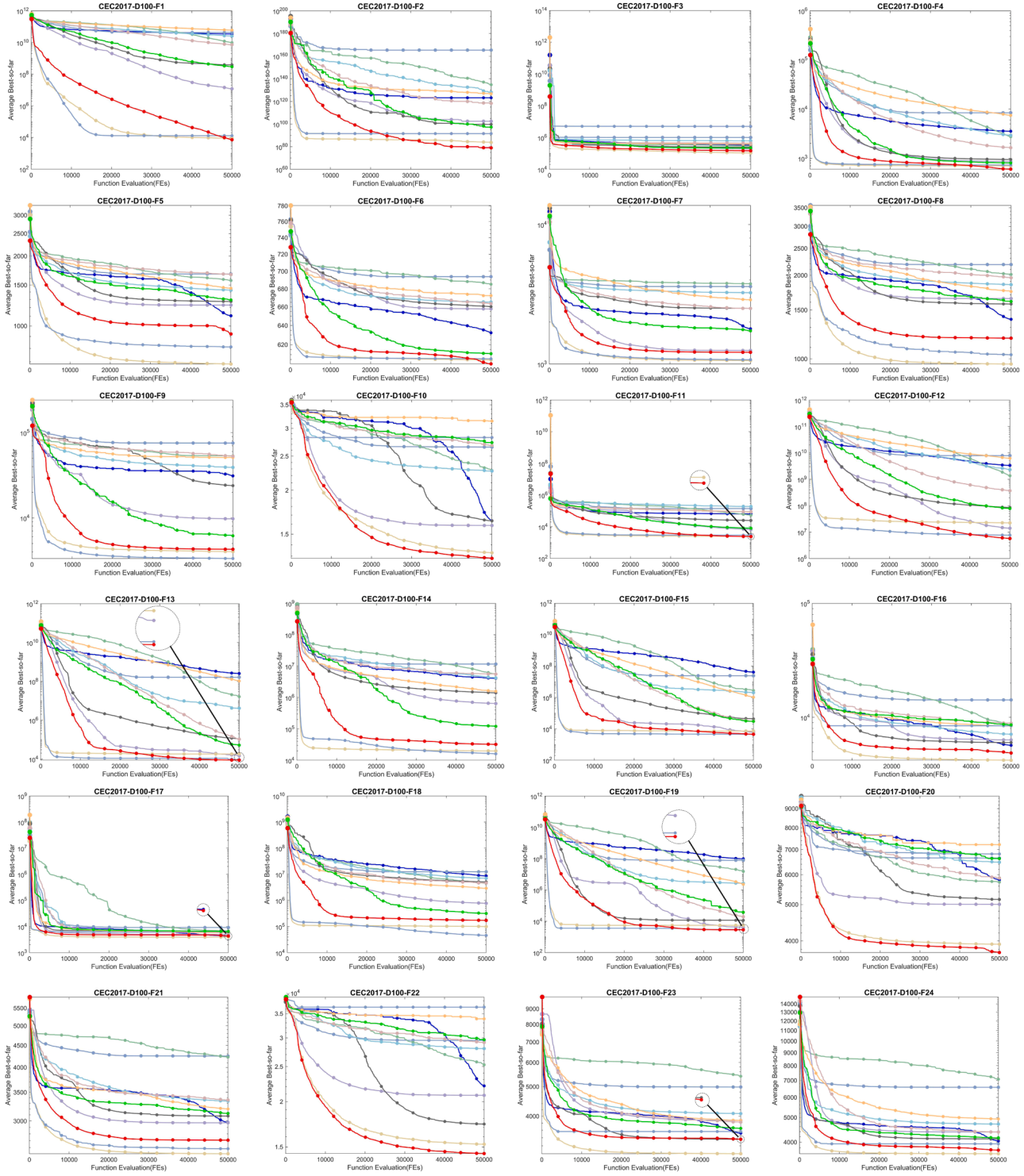


Fig. 6. Convergence curves of different algorithms ($D = 100$).

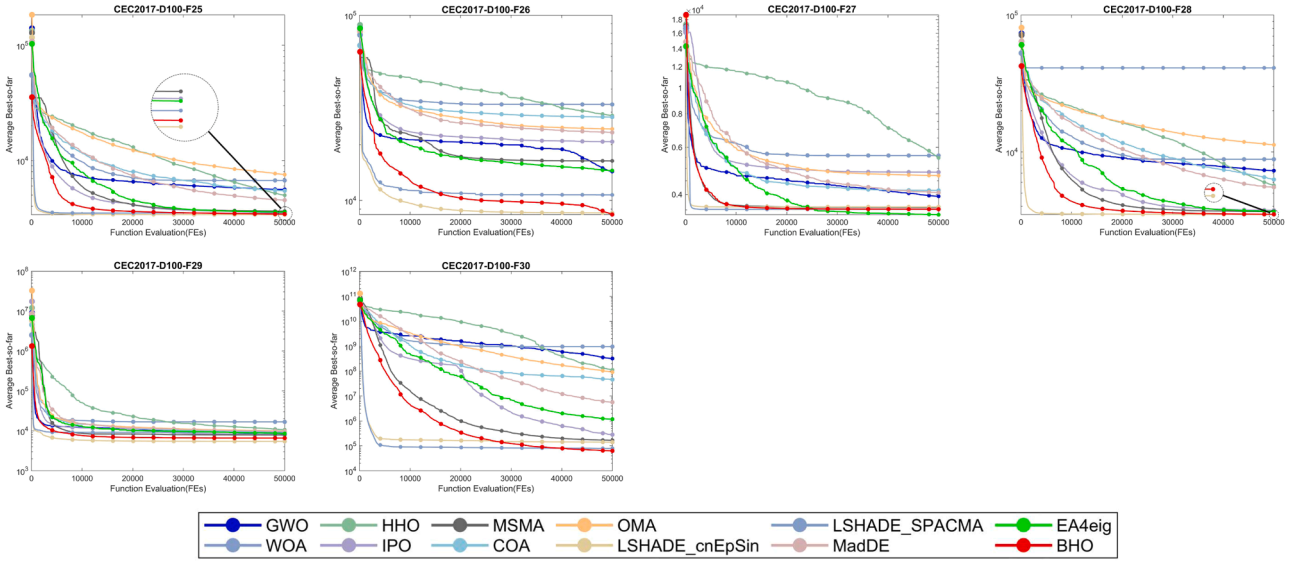


Fig. 6. (*continued*).

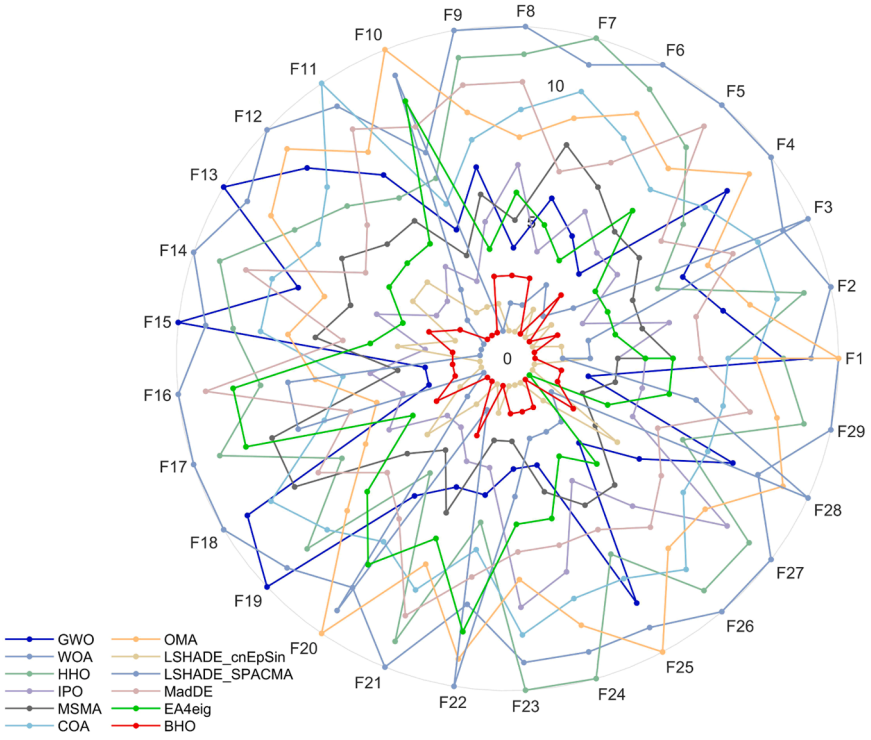


Fig. 7. Ranking distribution of different algorithms ($D = 100$).

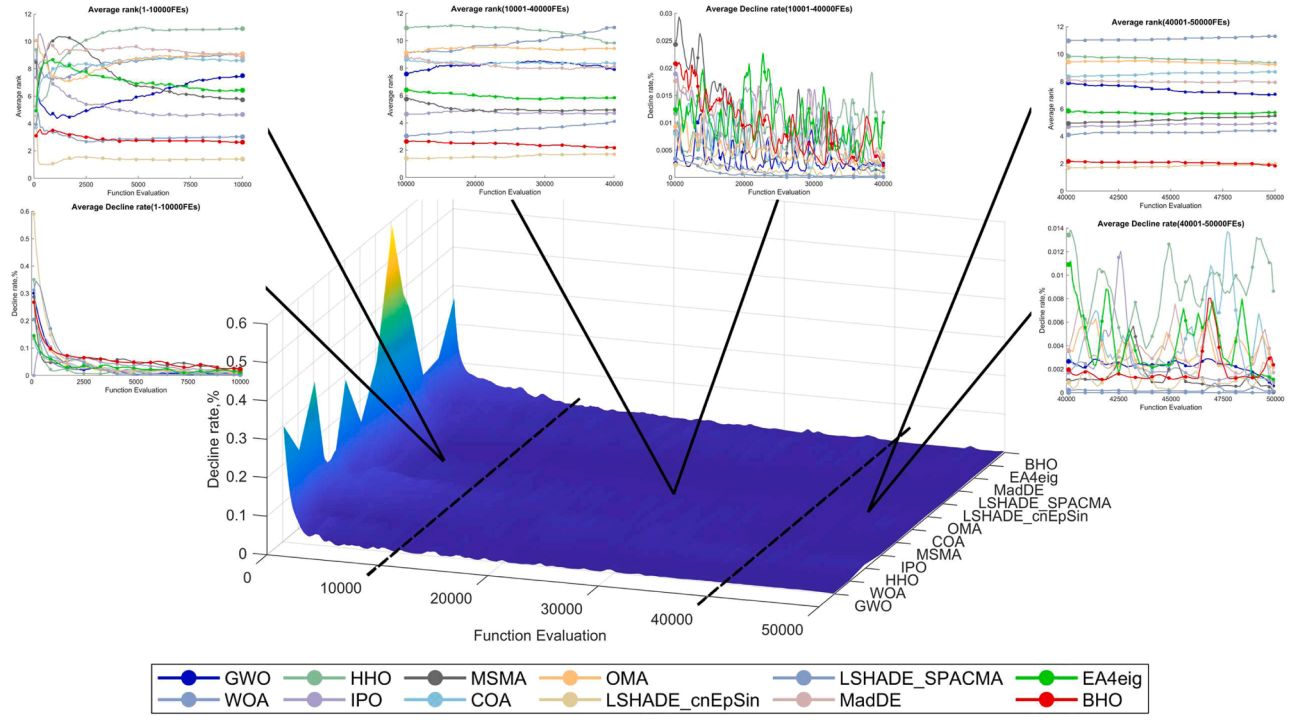


Fig. 8. Decline rate of different algorithms ($D = 100$).

Table 8
Set and parameter definitions.

Set	Definition	Parameter	Definition
P_k	Coordinates of terminal node k	d_{ij}	Distance from node i to node j
u_k	Remaining demand of terminal node k	ϑ	UAV service rate per unit time
O_{lv}	Coordinates of obstacle node lv	σ_o	Gaussian function variance of obstacle node o
E	Set of nodes in the network, $\{P, O\} \in E$	q	Safety distance between UAVs
A	Directed arc formed by two nodes	h	Communication distance
K	Set of terminals	η, β	Related parameters for the Sigmoid function of terminal demand
O	Set of obstacles	μ^t_{gk}	Service relationship between terminal k and UAV g at time t
G	Set of UAVs	p^t_g	Location of UAV g at time t
T	Set of time nodes	t_e	Planning termination time
W, L	Map size	w_d, w_r, w_u	Weight coefficients for path length, risk, and reward

servers to enhance services for end-users. Proper path planning ensures service quality while improving system stability, reliability, and computational efficiency [22,52,53]. This section establishes a rigorous mathematical model for Multi-UAV MEC-PP. Based on this model, we evaluate BHO's performance across various problem scales.

5.1. Mathematical modeling of multi-uav mec-pp

In the Multi-UAV MEC-PP problem, each UAV needs to plan its path reasonably to meet the end-user's demand with the shortest movement cost and minimal exposure risk. To facilitate the establishment of the mathematical model, the following assumptions are made for the specific scenario:

(1) Obstacle risk is modeled by an (unnormalized) isotropic Gaussian centered at each obstacle location with variance σ_o, where $\sigma_o = \{12, 5, 5, 4, 6, 6, 1, 12, 4, 10\}$ for the 10 obstacles;
(2) A Sigmoid function is used to describe the degree of satisfaction of the end-user demand;
(3) Other UAVs are treated as obstacles with Gaussian functions of the same variance for risk avoidance;

On this basis, the paper will define the related sets, parameters, and decision variables to characterize the constraint relationships and optimization objectives of Multi-UAV MEC-PP. Table 8 summarizes the set definitions and objective definitions. To keep the proposed engineering case focused on joint path planning and service scheduling under QoS constraints, we adopt a workload-normalized MEC abstraction. Specifically, each terminal k is associated with a remaining workload (or service demand u_k, and each UAV provides an effective service capability ε per unit time, which can be interpreted as an aggregated rate that implicitly accounts for both communication support and onboard edge processing. Under this abstraction, "load balance" in this paper refers to balancing the served workload across UAVs (rather than merely balancing the number of visited nodes). A more detailed MEC formulation explicitly modeling task input size (bits), CPU frequency/cycles, latency decomposition, and energy consumption is beyond the scope of this paper and will be considered in future work.

The decision variables are as follows:

$$X^t_{ijg} = \begin{cases} 1, & \text{if UAV } g \text{ traverse the directed arc}(i,j) \text{ at time } t \\ 0, & \text{otherwise} \end{cases} \tag{21}$$

The objective function is as follows:

$$\min Z = w_d C_d + w_r C_r + w_u C_u + (\max(C_u) - \min(C_u)) \tag{22}$$

$$C_d = \sum_{t \in T} \sum_{(i,j) \in A} \sum_{g \in G} d_{ij} X^t_{ijg} \tag{23}$$

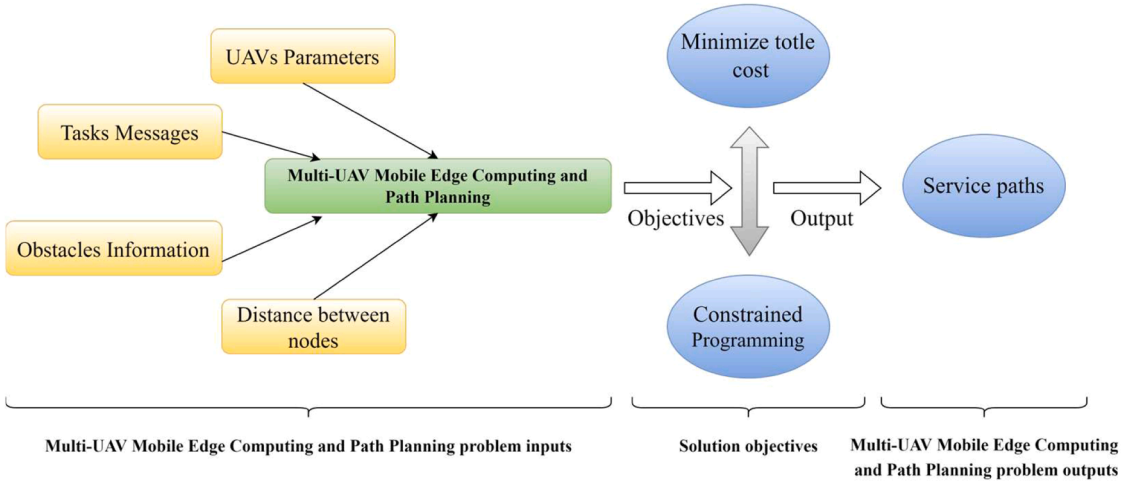


Fig. 10. Multi-UAV MEC-PP flow chart.

Table 9
Parameters related to Multi-UAV MEC-PP simulation.

Parameters	Value	Parameters	Value
$(W,\ L)_{\text{small}}$	60, 80	w_d	1
$(W,\ L)_{\text{Large}}$	100, 100	w_r	1
ε	0.2	w_u	1
q	2	β	2
η	5	h	11

$$C_r = \sum_{t\in T}\sum_{(i,j)\in A}\sum_{g\in G} X_{ijg}^{t}\left\{1-\prod_{o\in O}\left[1-e^{-\frac{d_{oi}^2}{2\sigma_o}}\right]\right\} \tag{24}$$

$$C_u = \sum_{t\in T}\sum_{(i,j)\in A}\sum_{g\in G}\sum_{k\in K}\mu_{gk}^{t}\left\{\exp\left[-\frac{u_k^{\eta}}{u_k+\beta}\right]-\exp\left[-\frac{(u_k-\vartheta)^{\eta}}{u_k+\beta}\right]\right\} \tag{25}$$

The constraints are as follows:

$$\mu_{gk}^{t}=\begin{cases}1 & d_{p_g^t p_k^t}\le h, \forall t\in T, \forall g\in G, \forall k\in K\\ 0 & d_{p_g^t p_k^t} > h, \forall t\in T, \forall g\in G, \forall k\in K\end{cases} \tag{26}$$

$$\sum_{t\in T} X_{ijg}^{t}\le 1\ , \forall (i,j)\in A, \forall g\in G \tag{27}$$

$$u_k^{t_e}\le 0.2, \forall k\in K \tag{28}$$

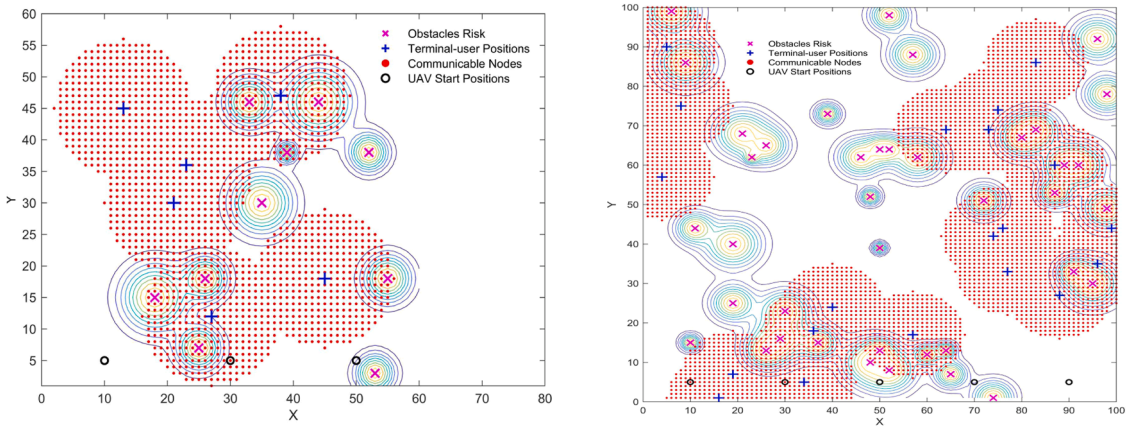
$$\sum_{g\in G}\sum_{g'\in G, g'\neq g} d_{p_g^t p_{g'}^t} > q, \forall t\in T \tag{29}$$

$$X_{ijg}^{t}=\{0,1\},\ \forall (i,j)\in A, \forall g\in G, \forall t\in T \tag{30}$$

The objective function (22) of the Multi-UAV MEC-PP established in this study aims to minimize the combined cost of path, risk, and workload deviation, while maximizing service quality. We quantify service

Table 10
Optimization results of a small-scale Multi-UAV MEC-PP case study.

Alogorithms	Best	Worst	Mean	Std	Rank
GWO	2.8672	3.8638	3.2242	0.4042	6
WOA	4.6076	5.7003	5.2439	0.4979	12
HHO	4.1623	7.5425	5.2362	1.3553	11
IPO	4.0476	7.1984	5.7300	1.2057	10
MSMA	3.3209	4.6604	3.9541	0.4837	7
COA	3.9031	4.8330	4.5379	**0.3966**	9
OMA	3.8038	5.3988	4.4110	0.6453	8
LSHADE_cnEpSin	2.8600	4.0091	3.5503	0.4619	5
LSHADE _SPACMA	1.9546	3.4866	2.5691	0.5591	2
MadDE	2.6560	4.1590	3.5285	0.6057	4
EA4eig	2.6340	3.7950	3.4977	0.4348	3
BHO	**1.8143**	**2.6799**	**2.2936**	0.4001	**1**



(a) Small-scale example map (b) Large-scale example map

Fig. 11. Distribution information of end users and obstacles.

Table 11
Objective function value of a small-scale Multi-UAV MEC-PP case study.

Algorithms	Z	C_d	C_r	C_u
GWO	2.8672	**86.8406**	1.5239	5.9977
WOA	4.6076	110.9706	2.0580	**5.9980**
HHO	4.1623	110.4264	1.6400	**5.9980**
IPO	4.0476	110.0244	1.4417	5.9977
MSMA	3.3209	96.8406	1.4300	5.9977
COA	3.9031	144.4975	2.4685	**5.9980**
OMA	3.8038	120.6690	2.1334	**5.9980**
LSHADE_cnEpSin	2.8600	118.4264	1.4370	5.9978
LSHADE _SPACMA	1.9546	111.8995	2.2446	5.9977
MadDE	2.6560	122.4558	2.1289	5.9978
EA4eig	2.6340	100.5980	2.1021	5.9977
BHO	**1.8143**	118.4680	**1.0172**	5.9978

quality using a bounded utility (Eq. (25)). Here, a Sigmoid function normalizes satisfaction to [0, 1] and models the saturation effect of diminishing marginal gains [54]. This Sigmoid-based satisfaction is used as a soft utility proxy to couple service quality with other objective terms [55]. In standard MEC formulations, QoS is more commonly specified via explicit latency limits and/or minimum data-rate requirements [56]. Here, the workload deviation term characterizes the imbalance of cumulative served workload among UAVs over the planning horizon. To achieve this goal, the penalty function method is used in this study to implement the following constraints:

1. Constraint (26) ensures that a UAV can establish a service association with an end user only when they are within communication range. Here, the threshold h is used as a simplified coverage-feasibility parameter, representing an effective radius corresponding to a minimum link-quality requirement (e.g., minimum SNR or achievable rate);
2. Constraint (27) ensures that each UAV g traverses a given arc (i, j) at most once over the planning horizon (across all time slots), thereby preventing redundant revisits/loops along the same arc.;
3. Constraint (28) ensures that the service quality for each terminal meets the preset standards and satisfies customer demand;
4. Constraint (29) ensures that UAVs always maintain a safe distance from each other;
5. Constraint (30) defines the range and characteristics of the decision variables, ensuring that the solution of the model is feasible both logically and numerically.

The features of Multi-UAV MEC-PP are shown in Fig. 10. By processing the system parameters and network information, the optimization algorithm minimizes the objective function (20) subject to constraints (26)–(30). This procedure yields the optimal task allocation and flight trajectories.

The proposed multi-UAV MEC-PP scenario is depicted in 2D for clarity and ease of presentation. This formulation extends naturally to 3D by defining UAV positions as (x, y, z) and using 3D distances. Altitude and vertical-rate constraints can also be incorporated as needed. Moreover, obstacle avoidance can be generalized from 2D regions to 3D volumes.

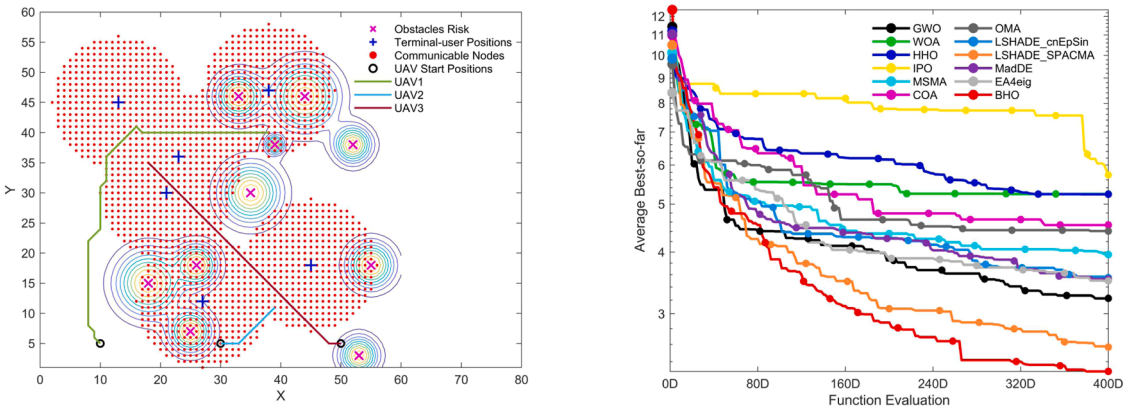
Section 5.1 presents a simplified yet reproducible UAV-assisted service optimization case study for evaluating the proposed optimizer under coupled objectives and constraints. For model tractability, communication feasibility is abstracted using a distance-threshold coverage rule, and service quality is represented using a bounded satisfaction/utility mapping to enable multi-term objective coupling. We note that a more comprehensive MEC formulation would explicitly

Table 12
Optimization results of a large-scale Multi-UAV MEC-PP case study.

Alogorithms	Best	Worst	Mean	Std	Rank
GWO	−3.0167	−0.4280	−1.7593	1.7781	5
WOA	1.8400	6.7521	4.2960	3.4734	12
HHO	1.6990	2.4673	2.0831	0.5433	10
IPO	3.8671	4.5885	4.2278	0.5101	11
MSMA	−2.6956	−1.7279	−2.2117	0.6843	4
COA	0.0775	0.5695	0.3235	0.3479	8
OMA	−0.9579	1.7366	0.3893	1.9053	9
LSHADE_cnEpSin	−2.8007	−0.3476	−1.5741	1.7346	6
LSHADE _SPACMA	−4.3822	−3.5821	−3.9822	0.5658	2
MadDE	−3.8819	−3.7475	−3.8147	**0.0950**	3
EA4eig	−0.5020	0.1814	−0.5240	0.9976	7
BHO	**−5.1929**	**−4.8825**	**−5.0377**	0.2195	**1**

Table 13
Objective function value of a large-scale Multi-UAV MEC-PP case study.

Algorithms	Z	C_d	C_r	C_u
GWO	−3.0167	421.2670	4.0593	**19.9933**
WOA	1.8400	735.0732	4.1262	**19.9933**
HHO	1.6990	732.8549	5.0000	**19.9933**
IPO	3.8671	557.9899	3.7462	**19.9933**
MSMA	−2.6956	417.8823	4.0000	**19.9933**
COA	0.0775	586.7351	4.9728	**19.9933**
OMA	−0.9579	457.7056	4.9998	19.9931
LSHADE_cnEpSin	−2.8007	440.6102	4.0444	**19.9933**
LSHADE _SPACMA	−4.3822	474.1493	5.0000	**19.9933**
MadDE	−3.8819	406.2498	4.0008	19.9931
EA4eig	−0.5020	513.4924	5.0000	**19.9933**
BHO	**−5.1929**	**352.1665**	**3.5141**	**19.9933**



(a) Optimal path planning result

(b) Iterative convergence curve of algorithms

Fig. 12. Small-Scale Simulation of Multi-UAV MEC-PP based on BHO.

Algorithm 1
QPRS.

Set the Number of Selected Teams $N_s = 0$
Set the Number of Unselected Teams $N_{us} = N_{group}$
Set the Selection Status of the Team $\varphi_l = 0$
Set $guard = 0$, **Set** $guard_{MAX}$.
while (N_s ! = 1) and ($guard < guard_{MAX}$) **do**
  if (N_s = = 0) **then**
    for (l in N_{group} and φ_l == 0) **do**
      if ($r > \alpha_l^2$) **then**
        Update the Selection Status of the Team $\varphi_l = 1$
        Update the Number of Selected Teams $N_s = N_s + 1$
        end if
      end for
  else
    for (l in N_{group} and φ_l == 1) **do**
      if ($r \le \alpha_l^2$) **then**
        Update the Selection Status of the Team $\varphi_l = 0$
        Update the Number of Selected Teams $N_s = N_s - 1$
        end if
    end for
  end if
  $guard = guard + 1$
end while
if ($guard \ge guard_{MAX}$) and ($\sum_l \varphi_l$! = 1) **then**
  Randomly select one index $l^* \in [1, N_{group}]$
  Set $\phi_{l^*} = 1$, $\phi_{h \ne l^*} = 0$; $N_s = = 1$
end if

model task input sizes and computational workloads, CPU-cycle demands and CPU-frequency allocation, computation latency and energy consumption. It would also adopt rate-/SNR-based air-to-ground communication models accounting for path loss, fading, and interference, together with latency- or rate-driven QoS constraints. Extending the present case study to incorporate these MEC components and performing a full 3D evaluation are left for future work.

5.2. Simulation experiment

This section evaluates the optimization performance of the proposed BHO in both small-scale and large-scale Multi-UAV MEC-PP scenarios. The small-scale setting consists of 10 obstacle nodes, 3 UAVs, and 6 terminal nodes, whereas the large-scale setting consists of 40 obstacle nodes, 5 UAVs, and 20 terminal nodes. The parameter values reported in Table 8 are chosen based on two principles: (i) consistency with the modeling assumptions in Section 5.1, and (ii) scale alignment with the discretized scenario size (W, L). Specifically, the communication threshold h is set according to the adopted distance-threshold association rule, serving as an effective coverage radius to keep service association feasible but non-trivial under the considered map scale. The separation distance q is selected as a conservative safety margin to support collision avoidance under discretized motion. The Sigmoid parameters (η, β) are configured to produce a smooth bounded satisfaction mapping as a normalized QoS proxy. Finally, the weights (w_d, w_r, w_u) are set to balance the magnitudes of the path-length, risk, and utility terms so that no single component dominates the composite objective under the same scale. The simulation parameters are summarized in Table 9, and the spatial distribution of terminal users, obstacles, and UAV initial positions is illustrated in Fig. 11.

5.2.1. Optimization performance evaluation under the small-scale multi-UAV MEC-PP model

This section conducts simulation comparison experiments for the small-scale Multi-UAV MEC-PP model. All comparison algorithms are run independently 5 times, with the average results used as the final ranking criterion. Table 10 presents the performance evaluation based on statistical indicators, including the best value, worst value, mean, and standard deviation. For clarity, the best results are highlighted in boldface. The objective function values for each algorithm are listed in Table 11.

The results show that the BHO algorithm performs the best in this case: its optimal values and averages from 5 independent runs outperform those of other algorithms, with the best optimization value being 1.8143, a 7.18% improvement over the second-ranked LSHADE_SPACMA (1.9546). Its optimal path and convergence curve are shown in Fig. 12. Overall, the experimental results validate the excellent performance and engineering application potential of BHO in small-scale Multi-UAV MEC-PP.

5.2.2. Optimization performance evaluation of large-scale multi-UAV MEC-PP model

This section further tests the optimization performance of the large-scale Multi-UAV MEC-PP model. In this scenario, due to more complex environmental constraints and interaction factors, the problem scale significantly increases, and the optimization difficulty also rises. The experimental results show that the BHO algorithm still maintains excellent performance in large-scale cases. Its statistical indicators, including optimal value, worst value, standard deviation, and mean, are listed in Table 12 (with the optimal value in bold), and the corresponding objective function values for each algorithm are shown in Table 13.

In the overall comparison, BHO ranks first in optimal value, worst value, and mean, with its relative advantage further expanding to 18.5% compared to the small-scale case. The optimal path results and convergence curve are shown in Fig. 13. This trend aligns with the CEC2017 suite results: as the problem dimensionality increases, BHO's performance advantage becomes more pronounced. This validates BHO's superiority in solving high-dimensional, complex engineering tasks.

6. Conclusion and future work

This paper proposes a novel MH algorithm–BHO, inspired by the intelligence gathering, task deployment, collaborative division of labor, and dynamic scheduling mechanisms exhibited by bounty hunter organizations in pursuit of targets. Unlike existing MH algorithms that rely on mean aggregation or centralized guidance, BHO introduces two core innovations in its algorithm design: First, the Explorpolis rule constructs a quantitative relationship model between individual fitness and solution space position, enabling intelligent screening and retention of inferior solutions. This balances exploitation and exploration at the population level while significantly enhancing diversity maintenance. Second, QPRS replaces the discrete sampling method of traditional roulette-wheel selection by utilizing the continuous evolutionary characteristics of probability amplitude, making the selection pressure smoother and more controllable, further reducing the risk of premature convergence.

In benchmark numerical optimization tests, BHO demonstrates excellent optimization performance on the CEC2017 test suite (30-dimensional and 100-dimensional), with its Friedman average ranking surpassing various mainstream and competition-winning algorithms. The statistical results based on the Wilcoxon rank-sum test also indicate that BHO significantly outperforms the comparison methods on most test functions, validating the statistical significance of its advantages. Furthermore, in the engineering application of Multi-UAV MEC-PP, BHO achieved optimal results in both small-scale and large-scale simulation environments. Its relative advantage further increases with the scale and complexity of the problem, fully demonstrating its applicability and generalization ability in high-dimensional complex optimization tasks.

Overall, BHO demonstrates strong global search and convergence capabilities on benchmark functions, while also showing broad application prospects in practical engineering optimization. Future research will focus on the following directions. First, further improving the algorithm's stability across multiple runs and reducing the variance of

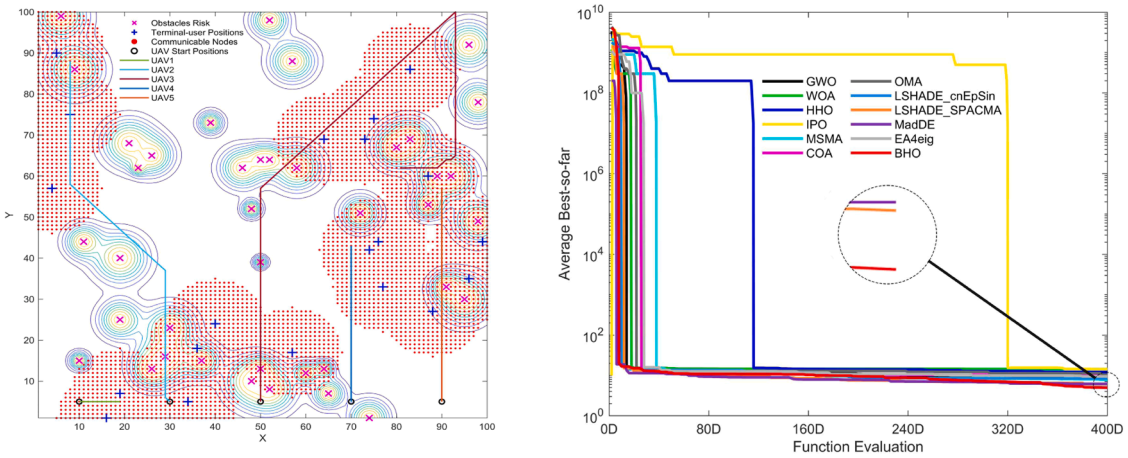
Algorithm 2
Proposed BHO.

Initialize population size N, maximum evaluations k_{max}, dimension D, define objective function $f(\cdot)$
Initialize BHO parameters N_{group}, ρ, L, U, c_1^0, c_2^0, c_3^0, r_h^1, $k = 1$, $C^0 = 0$, $MPPC$
Sample $X_o \in \mathrm{R}^{10N\times D}$ uniformly from $[L, U]$
Cluster X_o using the k-means algorithm to obtain centroid population $X(N \times D)$
Cluster X into N_{group} groups using *k-means*; assign each hunter to a group.
Calculate objective function $f(\cdot)$ for each hunter X_i^k as F_i^k
Set each hunter poor performance count $P_i = 0$
while $k < k_{max}$ **do**
    Update each hunter current value c_i^k as $c_i^k = 1 - \frac{F_i^k}{\sum F_i^k}$
    Update each group current value $C^{k,l}$ as $C^{k,l} = (1-\rho) * C^{k-1,l} + \sum_{i\in B} c_i^k$
    Update personal best position X_i^{best}, group best position $X_{best}^{k,l}$, and global best position X_{best}^k.
    Update each hunter ranking ratio R_{hi}^k
    Update each hunter movement distance m_i^k as $m_i^k = 1 - \frac{k}{2k_{max}} * \frac{F_i^k}{\sum F_i^k}$
    for ($i = 1$: N) **do**
        Calculate $\overline{c_i^k} = \frac{c_{1_i}^k + c_{2_i}^k + c_{3_i}^k}{3}$
    for ($j = 1$: N) **do**
        if (($R_{hi}^k \le 0.2$ & $c_j^k \ge \overline{c_i^k}$) || ($R_{hi}^k > 0.8$ & $c_j^k < \overline{c_i^k}$)) **then**
    Update reference factor c_j^k as $c_j^k = \min\left(c_j^{k-1} + \frac{R_{hi}^k}{10}, 0.8\right)$
        elseif (($R_{hi}^k \le 0.2$ & $c_j^k < \overline{c_i^k}$) || ($R_{hi}^k > 0.8$ & $c_j^k \ge \overline{c_i^k}$)) **then**
            Update reference factor c_j^k as $c_j^k = \max\left(c_j^{k-1} - \frac{R_{hi}^k}{10}, 0.2\right)$
        end if
        if ($R_{hi}^k > 0.8$) **then**
            Set $P_i = P_i + 1$
        if ($P_i = MPPC$) **then**
            Measure a new group based on $C^{k,l}$ and **assign** the hunter to this Group
        end if
        end if
        if ($R_{hi}^k \le r_h^k$ || P_i=$MPPC$) **then**
            Set $P_i = 0$
        Calculate test position of current hunter as $X_{i,j}^k test = X_{i,j}^k old + m_i^k * \left(c_{1_i}^k * rand * \left(X_{best,j}^k - X_{i,j}^k\right) + c_{2_i}^k * rand * \left(X_{best,j}^{k,team_l} - X_{i,j}^k\right) + c_{3_i}^k * rand * \left(X_{i,j}^{best} - X_{i,j}^k\right)\right)$
        else
          if ($F_r^k \ge F_i^k$) **then**
            Calculate mid position of current hunter as
$X_{i,j}^k rough = X_{i,j}^k + rand * \left(X_{r,j}^{k,team_l} - X_{i,j}^k\right)$
        else
            Calculate mid position of current hunter as
$X_{i,j}^k rough = X_{i,j}^k - rand * \left(X_{r,j}^{k,team_l} - X_{i,j}^k\right)$
        end if
            if ($rand_1 > 0.5$) **then**
              Calculate test position of current hunter as
                $X_{i,j}^k test = X_{i,j}^k rough + m_i^k * rand_1 * X_{i,j}^k rough$
            else
              Calculate test position of current hunter as
                $X_{i,j}^k test = X_{i,j}^k rough - m_i^k * rand_1 * X_{i,j}^k rough$
            end if
    end if
    Calculate objective function f_n for each hunter $X_i^k test$ as $F_i^k test$
    for ($i = 1$: N) **do**
      if ($F_i^k test > F_i^k$) **then**
            Update position of current hunter as $X_i^{k+1} = X_i^k test$, $F_i^{k+1} = F_i^k test$
        else
            Calculate the distance TD_i between each solution and the optimal solution as $TD_i^k = \frac{\sqrt{\sum_{j=1}^{D}\left(\frac{X_{new,j}^{k+1} - X_{i,j}^k}{U_j - L_j}\right)^2}}{D}$
            if (any $F > F_i test \& TD > TD_i$) **then**
                Update position of current hunter as $X_i^{k+1} = X_i^k$, $F_i^{k+1} = F_i^k$
            else
                Update position of current hunter as $X_i^{k+1} = X_i^k test$, $F_i^{k+1} = F_i^k test$
              end if
        end if
        if ($F_i^{k+1} > F_i^k$)
            Update hunters updated score as $score_i^k = \frac{1}{N * k_{max}}$
        else

(*continued on next page*)

Algorithm 2 (*continued*)

Update hunters updated score as $score_i^k = \frac{-1}{N * k_{max}}$
end if
end if
Update threshold r_h for hunter behavior patterns as $r_h^{k+1} = r_h^k + \sum_{i \in M_t} score_i^{k-1} - \sum_{i \in M_r} score_i^{k-1}$
end while
Return best hunter X_{best}^k



(a) Optimal path planning result (b) Iterative convergence curve of algorithms

Fig. 13. Large-Scale Simulation of Multi-UAV MEC-PP based on BHO.

results. Second, extending BHO to multi-objective and dynamic optimization scenarios to achieve more efficient approximation of Pareto optimal sets. Third, combining distributed computing and parallel mechanisms to improve operational efficiency in large-scale problems. (The source code of BHO will be made available upon publication.)

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Data availability

Matlab codes of BHO are available at https://ww2.mathworks.cn/matlabcentral/fileexchange/183483-bounty-hunter-optimizer.

Ethical approval

This article does not contain any studies with human participants or animals performed by any of the authors.

CRediT authorship contribution statement

Mingyang Yu: Writing – review & editing, Writing – original draft, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Haorui Yang:** Writing – review & editing, Writing – original draft, Software. **Jiaqi Zhang:** Software, Resources, Conceptualization. **Kaichen Ouyang:** Visualization, Software, Resources, Methodology. **Shengwei Fu:** Writing – review & editing, Visualization, Formal analysis. **Panlong Tan:** Supervision, Resources, Formal analysis. **Frank Jiang:** Writing – review & editing, Visualization, Resources. **Jing Xu:** Supervision, Resources, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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